

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM



CURRICULUM AND SYLLABUS FOR THE COURSES OF B.TECH /B.TECH (HON)/DUAL DEGREE (B.TECH - MS) PROGRAMME IN ELECTRONICS AND COMMUNICATION ENGINEERING

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B.Tech./B.Tech (Hon.)/ B.Tech - MS General Course Structure

Electronics and Communications Engineering

Semester - I						Semester - II					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA111	Discrete Mathematics	3	1	0	4	IMA121	Calculus and Linear Algebra	3	1	0	4
ICS112	Computer Programming	3	1	2	5	ICS121	Data Structures I	3	1	2	5
IEC111	Electronic Circuits	3	1	2	5	ICS122	Computer Organization	3	1	0	4
IEC112	Network Theory	3	1	0	4	IEC121	Digital Design and Electric Circuits	3	1	2	5
IHS111	Communication Skills	3	0	0	3	IEC122	Signals and Systems	3	1	0	4
IHS112	Foreign Language	1	0	0	1	IHS121	Personality Development	1	0	0	1
IPT111	Physical Training I	0	0	2	0						
Total		16	4	6	22	Total		16	5	4	23
Cumulative Credits at the End of First Year: 45											
Semester - III						Semester - IV					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA211	Probability, Statistics, and Random Processes	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
IEC211	Solid State Devices (ADM 2024 onwards)	3	0	0	3	IEC221	Digital Design with Hardware Description Languages	3	0	2	4
IEC212	Analog Electronics	3	0	2	4	IEC222	Digital Communication	3	0	2	4
IEC213	Microprocessors & Microcontrollers	3	0	2	4	IEC223	Electromagnetic Theory	3	1	0	4
IEC214	Principles of Communication	3	1	0	4	ICS224	Computer Networks	3	0	2	4
IEC215	Linear Integrated Circuits	3	1	0	4	ICS225	Data Structures III	1	0	2	2
ICS215	Data Structures II	1	0	2	2	IHS 221	Fundamentals of Economics	1	0	0	1
IPT211	Physical Training II	0	0	2	0	IHS222	Principles of Management	1	0	0	1
Total		19	3	8	25	Total		18	2	8	24
Cumulative Credits at the End of Second Year: 94											
Semester - V						Semester - VI					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
ECEXXX	Bouquet Core I	3	0	2	4	ECEXXX	Bouquet Core III	3	0	2	4
ECEXXX	Bouquet Core II	3	0	2	4	ECEXXX	Bouquet Core IV	3	0	2	4
ECSXXX	EC Elective I	3	0	2	4	ECSXXX	EC Elective II	3	0	0	3
IEGXXX	Engineering Elective	3	1	0	4	ECSXXX	EC Elective III	3	1	0	4
IMAXXX	Mathematics Elective	3	0	0	3	IOEXXXX	Science Elective	3	0	0	3
IHSXXX	Humanities Elective I	1	0	0	1	IOEXXXX	Institute Open Elective I	3	0	0	3
IHSXXX	Humanities Elective II	1	0	0	1	IHP321	Honours Project Phase I (Optional) I				
IHSXXX	Humanities Elective III	1	0	0	1						
Total		18	1	6	22	Total		18	1	4	21
Cumulative Credits at the End of Third Year: 137											
Semester - VII						Semester - VIII					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
ECEXXX	Bouquet Core V	3	0	2	4	ECSXXX	EC Elective V	3	0	0	3
ECEXXX	Bouquet Core VI	3	0	2	4	ECSXXX	EC Elective VI/Industrial Training	3	1	0	4
ECSXXX	EC Elective IV	3	1	0	4	IOEXXXX	Institute Open Elective III	3	0	0	3
IOEXXXX	Institute Open Elective II	3	0	0	3	IBP421	B.Tech Project Phase II	6	0	0	6
IBP411	B.Tech Project Phase I	6	0	0	6	IHP422	Honours Project Phase II (Optional)				
ECSXXX	Research Course (Optional)					ECSXXX	Research Course (Optional)				
Total		18	1	4	21	Total		15	1	0	16
Cumulative Credits at the End of Fourth Year: 174 (B.Tech); 174+12=186 (B.Tech(Hon.)); 174+20=194(B.Tech – MS)											
Semester-IX						Semester -X					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IRP511	Research Project	12	0	0	12	IRP521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (BTech-MS)											

B.Tech./B.Tech (Hon.)/ B.Tech - MS Detailed Course Structure

Electronics and Communications Engineering

Semester - I					Semester - II						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA111	Discrete Mathematics	3	1	0	4	IMA121	Calculus and Linear Algebra	3	1	0	4
IEC111	Electronic Circuits	3	1	2	5	IEC121	Digital Design and Electric Circuits	3	1	2	5
ICS112	Computer Programming	3	1	2	5	ICS121	Data Structures I	3	1	2	5
IEC112	Network Theory	3	1	0	4	ICS122	Computer Organization	3	1	0	4
IHS111	Communication Skills	3	0	0	3	IEC122	Signals and Systems	3	1	0	4
IHS112	Foreign Language	1	0	0	1	IHS121	Personality Development	1	0	0	1
IPT111	Physical Training I	0	0	2	0						
Total		16	4	6	22	Total		16	5	4	23
Cumulative Credits at the End of First Year: 45											
Semester - III					Semester - IV						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA211	Probability, Statistics, and Random Processes	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
IEC211	Solid State Devices(ADM 2024 onwards)	3	0	0	3	IEC221	Digital Design with Hardware Description Languages	3	0	2	4
IEC212	Analog Electronics	3	0	2	4	IEC222	Digital Communication	3	0	2	4
IEC213	Microprocessors and Microcontrollers	3	0	2	4	IEC223	Electromagnetic Theory	3	1	0	4
IEC214	Principles of Communication	3	1	0	4	ICS224	Computer Networks	3	0	2	4
IEC215	Linear Integrated Circuits	3	1	0	4	ICS225	Data structures III	1	0	2	2
ICS215	Data Structures II	1	0	2	2	IHS221	Fundamentals of Economics	1	0	0	1
IPT211	Physical Training II	0	0	2	0	IHS222	Principles of Management	1	0	0	1
Total		19	3	8	25	Total		18	2	8	24
Cumulative Credits at the End of Second Year: 94											
Semester - V					Semester - VI						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSE311	Artificial Intelligence	3	0	2	4	ECE323	Internet of Things	3	0	2	4
ECS313	VLSI Design	3	0	2	4	ECE322	Embedded Systems	3	0	2	4
IEG314	Digital Signal Processing and its Applications	3	0	2	4	IOEXXX	Advanced Computer Architectures/ Computer Vision / Wireless Sensor Networks/ Cloud Computing/ System-On-Chip Design	3	0	0	3
IMA312	Information Theory and Coding	3	0	0	3	ECSXXX	(ADM 2024 onwards)ASIC Design/ Advanced Communication Networks/ Automotive Electronics/ Micro-Electro-Mechanical-Systems/ VLSI Testing and Verification Biomedical Signal Processing /Advanced Digital Signal Processing	3	1	0	4
ECE312	Microwave Engineering	3	1	0	4	ICS322	Machine Learning	3	0	0	3
IHS311	Human Resource Management	1	0	0	1	ISC321	High performance and Scientific Computing Control Systems	3	0	0	3
IHS312	Financial Management and Accounting	1	0	0	1	IHP321	Honours Project Phase I (Optional)				
IHS313	Operations and Supply Chain Management	1	0	0	1						
Total		18	1	6	22	Total		18	1	4	21
Cumulative Credits at the End of Third Year: 137											
Semester - VII					Semester - VIII						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
ECE411	FPGA Based System Design	3	0	2	4	ECSXXX	Data Compression/Design of Digital Integrated Circuits /Natural Language Processing/Realtime Embedded Systems /Optoelectronics/ Mixed Signal Design	3	0	0	3
ECE412	Antenna Theory and Design	3	0	2	4	ECSXXX	Nanoelectronics/ Multimedia Communication/ Speech and Audio Processing Radar Engineering/ Robotics and Automation / Industrial training/ Advanced Wireless Communication Techniques	3	1	0	4
ECSXXX	Statistical Signal Processing/ Optical Communication Networks /Mixed Signal Design/ System-On-Chip Design / Electronic Measurements and Instrumentation/ Wireless and Cellular Communication/ Nanoelectronics/ Analog IC Design	3	1	0	4	IOEXXX	Large Language Models/ Error Control Coding	3	0	0	3
IOEXXX	Cryptography and Network Security/ Optimization/ Detection and Estimation Theory / High Speed Networks/ Digital Image Processing/ Pattern Recognition	3	0	0	3	IBP422	B. Tech Project Phase II	6	0	0	6
IBP412	B. Tech Project Phase I	6	0	0	6	IHP422	Honours Project Phase II (Optional)				
ECSXXX	Research Course (Optional)					ECSXXX	Research Course (Optional)				
Total		18	1	4	21	Total		15	1	0	16
Cumulative Credits at the End of Fourth Year:174 (B. Tech); 174+12=186 (B. Tech(Hon)); 174+20=194 (B. Tech- MS))											
Semester - IX					Semester - X						
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IRP511	Research Project	12	0	0	12	IRP521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (B. Tech-MS)											

Remark: To meet the minimum requirement of 186 credits for qualifying the B.Tech (Hon) Degree, students may take two additional projects of 6 credits each and, to meet the requirement of 218 credits for B.Tech-MS, students may take two additional projects of 6 credits each, two 4 credit research courses and 24 credit research project in addition to 174 credits requirement of B.Tech Degree.

B.Tech./B.Tech (Hon.)/ B.Tech - MS Programme - Credit Requirement

Sl. No.	Course Description	Minimum Credits Requirement			Period
		B.Tech	B.Tech(Hon)	B.Tech-MS	
1	Institute Core/Foundation Courses	94	94	94	Semester I to IV
2	Bouquet Core Courses	24	24	24	Semester V to VIII
3	Stream Electives	18	18	18	Semester V to VIII
4	Institute Open Electives	9	9	9	Semester V to VIII
5	Humanities Electives	3	3	3	Semester V to VIII
6	Science Electives	3	3	3	Semester V to VIII
7	Maths Elective	3	3	3	Semester V to VIII
8	Engineering Elective	4	4	4	Semester V to VIII
9	Any other Electives/Industrial Training	4	4	4	Semester V to VIII
10	B.Tech Projects	12	12	12	Semester VII to VIII
11	Honours Project	-	12	12	Semester VI to VIII
12	Research Courses	-	-	8	Semester VI to VIII
13	Research Project	-	-	24	Semester IX to X
Total Credits required for Successful Completion		174	186	218	
Minimum CGPA required for Successful Completion		5.5	8.0	8.0	

SEMESTER I

IMA111 Discrete Mathematics [3-1-0-4]

Course Objectives

- To extend student's Logical and Mathematical maturity and ability to deal with abstraction
- To introduce most of the basic terminologies used in Electronics and Communication engineering courses.
- To explain and apply the basic methods of discrete mathematics in Electronics and Communication engineering.
- To able to write clear, concise and correct mathematics proofs.
- To solve counting problems involving permutations and combinations and apply Pigeon hole principle.
- To understand the basics of graph theory and group theory.

Course Outcomes

- Have knowledge of the concepts needed to test the logic of a program.
- Have an understanding in identifying structures on many levels.
- Be aware of a class of functions which transform a finite set into another finite set which relates to input and output functions in computer science.
- Be able to apply basic counting techniques to solve combinatorial problems
- Acquire ability to describe computer programs in a formal mathematical manner.

Syllabus

Logic: Propositions, negation, disjunction and conjunction, implication and equivalence, truth tables, predicates, quantifiers, rules of inference, methods of proof.

Set theory: definition and simple proofs in set theory, Inductive definition of sets and proof by induction, inclusion and exclusion principle, relations, representation of relations by graphs, properties of relations, equivalence relations and partitions, partial orderings, linear and well-ordered sets.

Functions: mappings, injection and surjections, composition of function, inverse functions, special functions, recursive function theory, Z-transform.

Elementary combinatorics: Counting techniques, pigeonhole principle, recurrence relation, generating functions.

Graph theory: Elements of graph theory, Euler graph, Hamiltonian path, trees, tree traversals, spanning trees.

Algebra: groups, Lagrange's theorem, homomorphism theorem, rings and fields, structure of the ring $\mathbb{Z}_n$ and the unit group $\mathbb{Z}_n^*$, lattice.

Textbooks/References

1. Kenneth H. Rosen, Discrete Mathematics and Its Applications, Seventh Edition, McGrawHill, 2017.

2. Norman L. Biggs, Discrete Mathematics, Oxford University Press, Second Edition, 2003.
3. J.P. Tremblay, R. Manohar, Discrete Mathematical Structures with applications to Computer Science, McGraw Hill, 2017.
4. K.A. Ross, C.R. B. Wright, Discrete Mathematics, 5th Edition, Pearson, 2003.
5. P. B. Bhattacharya, S. K. Jain, S. R. Nagpaul, Basic Abstract Algebra, Second Edition, Cambridge University Press, 2003.
6. J.A. Gallian, Contemporary Abstract Algebra, Ninth Edition, Cengage Learning, 2017.

IEC111 Electronic Circuits [3-1-2-5]

Course Objectives

- To impart the basic concepts of semiconductor devices
- To make students capable of analyzing the operation of various electronic circuits
- To develop knowledge for designing simple analog electronic circuits using discrete and integrated components.

Course Outcomes

- Ability to analyze PN junctions in semiconductor devices under various conditions.
- Ability to design and analyze simple rectifier/clipper/-clamper circuits using diodes
- Ability to design and analyze amplifier circuits using BJTs and MOSFETs
- Ability to simulate simple electronic circuits using softwares such as LTSpice/PSpice

Syllabus

Semiconductors: Intrinsic semiconductors, doped semiconductors, current flow in semiconductors, drift current, diffusion current, the pn junction, physical structure, semiconductor diode, diode equation, diode characteristics, static and dynamic resistances, diode equivalent models: ideal diode model, piecewiselinear model, constant voltage-drop model, diode applications: rectifiers, clipping and clamping circuits, zener diodes as regulators.

Bipolar Junction Transistors (BJTs): Introduction to transistors, transistor construction, operation, NPN and PNP transistors, transistor voltages and currents, transistor characteristics. Common-base and common-emitter transistor configurations, biasing BJTs: fixed-bias, emitter-stabilized bias, voltage-divider bias, dc bias with voltage feedback, emitter follower bias, current mirror, BJT as amplifiers, voltage gain, current gain, input resistance and output resistance using r_e model, BJT as a switch.

Field Effect Transistors: Introduction, MOSFET, device structure and physical operation, regions of operation, current-voltage characteristics, body effect, biasing of MOSFET. Optoelectronic devices: photodiodes - current and voltage in

an illuminated junction, photodetectors. Light Emitting Diode (LED).

Lab Practice

Familiarization of basic electronic lab equipments and components, diode characteristics, clippers, clampers, rectifiers, zener regulator, BJT input and output characteristics, MOSFET input and output characteristics, simulation of simple diode and transistor circuits.

Textbooks/References

1. Sedra A. and Smith K. C, "Microelectronic Circuits", Oxford University Press, Seventh Edition, 2015.
2. Robert. L. Boylestad, Louis Nashelsky, "Electronic Devices and Circuits Theory", Pearson Education, Eleventh Edition, 2015.
3. David A Bell, "Electronic Devices and Circuits", Oxford University Press, Fifth Edition, 2008.
4. Jacob Millman and Christos. C. Halkias, "Electronic Devices and Circuits", Mc. Graw Hill, Fourth Edition, 2015.
5. Ben. G. Streetman, Sanjay Kumar Banerjee, "Solid State Electronic Devices", 7th Edition, Pearson Education, 2016, ISBN 978-93-3255508- 2.
6. Donald A Neamen, Dhrubus Biswas, "Semiconductor Physics and Devices", 4th Edition, MCGraw Hill Education, 2012, ISBN 978-0-07-107010-2.
7. S. M. Sze, Kwok K. Ng, "Physics of Semiconductor Devices", 3rd Edition, Wiley, 2018

ICS112 Computer Programming [3-1-2-5]

Course Objectives

- To introduce the problem-solving processes/techniques.
- To teach computer programming
- To use C & C++ for solving the problems

Course Outcomes

- Students learn how to write the sequence of operations in solving a problem
- Students learn to translate the problem solving steps to a program
- Students learn the use of programming language for solving real world problems on a computer

Syllabus

Basics of computers: software/ systems, Programming- Introduction, Problem solving, Introduction, Problem solving techniques: definition of problems, solutions, top-down approach, breaking problem in to sub-problems.

Algorithms: writing the steps required, solving problems, representing algorithms as flowchart, translating to procedure/ functions, Modularity

Example problems: computation of factorial, sine, Mod arithmetic computation of quotient/ remainder, solving factorial

through recursion, etc. Object oriented technology- introduction, C ++ data types/ scope rules, C ++ control statements, Example problems/ program, Example problems/ program, Character handling, Pointers, functions, Classes and objects, Classes and objects.

Lab Practice

- Implement fundamental domain knowledge of the course for developing effective computing solutions by incorporating creativity and logical reasoning.
- Students are encouraged to use the lab sessions as a multi-use, technology-enhanced teaching space with characteristics of both classrooms and labs.
- Understand and learn how a big program can be broken up into independent modules and define functions and call them with appropriate parameters.
- Students should gain a clear idea of how decision making and various basic/advanced constructs for control flow and instruction repetition is done while programming.
- Students should learn how to use arrays for storing/retrieving large amount of data. They should also understand the concept of strings and string libraries used for their manipulation.
- Comprehend how to use structures as a compound datatype. Students should also acquire the capability to design structures according to their requirement.
- Understand recursion, pointer referencing/dereferencing and dynamic allocation of memory.

Textbooks/References

1. R G Dromey, "How to Solve It by Computer", Prentice-Hall International Series in Computer Science, 2006.
2. G. Michael Schneider, "Invitation to Computer Science", Eighth Edition, 2018.
3. Byron S Gotfried, "Programming with C", Third Edition, McGraw Hill Companies, 2017.
4. Michael Vine, "C Programming for the Absolute Beginner", Third Edition, 2014.
5. Brian W Kernighan, Dennis M. Ritchie, "C Programming Language", Second Edition, Pearson Education India, 2015.
6. Herbert Schildt, "C++ Complete Reference", McGraw Hill, Fourth Edition, 2017.
7. Eric Nagler, "Learning C++ : A hands-on Approach", Third Edition, Cengage learning, 2017.

IEC112 Network Theory [3-1-0-4]

Course Objectives

This course will enable students to:

- Describe basic network concepts emphasizing source transformation, source shifting, mesh and nodal techniques to solve for resistance, voltage, current and power.
- Explain Thevenin's, Millman's, Superposition, Maximum Power transfer and Norton's Theorems and apply them in solving the problems related to Electrical Circuits.
- Explain the behavior of networks subjected to transient conditions.
- Use applications of Laplace transforms to network problems.
- Study two port network parameters like Z, Y, T and h and their inter-relationships and applications.
- Study of RLC Series and parallel tuned circuit.

Course Outcomes

At the end of the course students will be able to

- Determine currents and voltages using source transformation/ source shifting/ mesh/ nodal analysis and reduce given network using star-delta transformation/source transformation/ source shifting.
- Solve network problems by applying Superposition/ Reciprocity/ Thevenin's/ Norton's/ Maximum Power Transfer/ Millman's Network Theorems and electrical laws to reduce circuit complexities and to arrive at feasible solutions.
- Calculate current and voltages for the given circuit under transient conditions.
- Apply Laplace transform to solve the given network
- Solve the given network using specified two port network parameter like Z or Y or T or h
- Understand the concept of resonance

Syllabus

Basic Concepts: Basics of electrical network element: resistance, inductance, capacitance, Equivalent resistance, inductance and capacitance, Energy stored in an inductance and capacitance, Practical sources, Source transformations, Kirchhoff's current and voltage laws Network reduction using Star – Delta transformation, Loop and node analysis with linearly dependent and independent sources for DC and AC networks.

Network Theorems: Superposition, Millman's theorems, Thevenin's and Norton's theorems, Maximum Power transfer theorem, Reciprocity theorem.

Transient behavior and initial conditions: Behavior of circuit elements under switching condition and their Representation, evaluation of initial and final conditions in RL, RC and RLC circuits for AC and DC excitations. Laplace Transformation & Applications: Solution of networks, step, ramp and impulse responses, waveform Synthesis.

Two port network parameters: Definition of Z, Y, h and Transmission parameters, modelling with these parameters, relationship between parameters sets.

Resonance: Series Resonance: Variation of Current and Voltage

with Frequency, Selectivity and Bandwidth, Q-Factor, Circuit Magnification Factor, Selectivity with Variable Capacitance, Selectivity with Variable Inductance.

Parallel Resonance: Selectivity and Bandwidth, Maximum Impedance Conditions with C, L and f Variable, current in Anti-Resonant Circuit, The General Case-Resistance Present in both Branches.

Textbooks/References

1. M.E. Van Valkenberg (2000), —Network analysis, Prentice Hall of India, 3rd edition, 2000, ISBN: 9780136110958.
2. Roy Choudhury, - Networks and systems, 2nd edition, New Age International Publications, 2006, ISBN: 9788122427677
3. Hayt, Kemmerly and Durbin, Engineering Circuit Analysis, TMH 7th Edition, 2010.
4. J. David Irwin /R. Mark Nelms, Basic Engineering Circuit Analysis, John Wiley, 8th ed, 2006.
5. Charles K Alexander and Mathew N O Sadiku, Fundamentals of Electric Circuits, Tata McGraw-Hill, 3rd Ed, 2009.

IHS111 Communication Skills [3-0-0-3]

Course Objectives

- The objective of the course is to improve the English communication skills of First Semester B. Tech students who had just passed out of their Senior Secondary classes. This is challenging since the class included students from various parts of the country speaking various mother tongues.
- The syllabus is designed to give importance to essential grammar, as well as reading, writing and speaking skills.
- Work in class consisted of teaching grammar, interspersed with written exercises, reading practice, reading comprehension, business letter writing, report writing, training in preparing CV for job applications, group discussion and extempore speaking.
- The classes will be rounded off with some training in the so called "soft skills".

Course Outcomes

At the end of the sessions:

- Improvement in English language ability was noted in most of the students.
- A large number showed very good improvement, while even the least competent registered some improvement.
- Under the circumstances, the objective of the class would appear to have been achieved.

Syllabus

Communication, verbal and non-verbal, Conversation: formal and informal, prepared and extempore, English: British/American/Indian, Vocabulary development: reading, use of dictionaries, Expression: writing, Pronunciation: phonetics, use of phonetic dictionaries, speaking, English grammar: Basics: Parts of speech: Noun, Pro-noun, Adjective, Verb, Adverb, Preposition, Conjunction, Interjection .Verb: Tenses. Sentence structure: S+V. Concord: Subject-Verb agreement.Reported speech,Active and passive voice, Tag questions, Confusing words and expressions,Synonyms and antonyms,Idioms and phrases, Common errors in English,Punctuation.Writing skills: Letters: Formal/Informal, Reports,CV. Comprehension: Listening/Reading/Making notes/ Summarising, Interview skills, Group discussion,Soft-skills.

Textbooks/References

1. A.J. Thomson & A.V. Martinet.A Practical English Grammar. Delhi: OUP.
2. George Yule.xford Practice Grammar: Advanced.Oxford:OUP.
3. Raymond Murphy.Essential English Grammar. Delhi: Cambridge University Press.
4. Mathew Monippally.The Craft of Business Letter Writing. New Delhi: Tata McGraw-Hill.

IHS112 Foreign Language [1-0-0-1]

Course Objectives

- To introduce the basics of the German language and its grammar to the students.
- To develop the four fundamental language skills at the initial level: listening, speaking, reading, and writing.
- To cover the essentials of the German language, such as the German alphabet and pronunciation, core grammar rules, and simple, everyday vocabulary.
- To provide a basic understanding of the German language and the culture of German-speaking countries (Germany, Austria, Switzerland, etc.).

Course Outcomes

Upon successful completion of this course, students will be able to:

- Introduce themselves and others, and ask and answer basic personal questions (e.g., about where they live, people they know, and things they have).
- Understand and use familiar, everyday expressions and very simple sentences aimed at satisfying concrete needs (e.g., ordering food, asking for directions).
- Interact in a simple way, provided the other person talks slowly and clearly and is prepared to help.
- Read and understand the main points in short, simple texts like public notices, posters, or short personal messages.

- Write a short, simple text, such as a postcard with holiday greetings, or fill in forms with personal details like their name, nationality, and address.
- Develop a basic awareness of German culture and daily life.

Syllabus

Introduction and Basics: The German Alphabet and Pronunciation (Das deutsche Alphabet und Aussprache): Learning the sounds of German letters and letter combinations. Greetings and Introductions (Begrüßungen und Vorstellungen): Formal and informal greetings, introducing oneself and others. Numbers, Time, and Dates (Zahlen, Uhrzeit und Datum): Numbers from 0 to 100, telling time, and mentioning dates.

Basic Grammar and Sentence Structure: Nouns, Gender, and Articles (Nomen, Genus und Artikel): Understanding that German nouns have a gender (masculine, feminine, neuter). Using definite and indefinite articles (*der/die/das, ein/eine/ein*) in the Nominative case. Simple Sentence Structure: Forming basic statement sentences and questions (W-Fragen like *Wer? Was? Wo?* and Ja/Nein questions). Verb Conjugation (Present Tense) (Konjugation im Präsens): Conjugating regular verbs and important irregular verbs like *sein* (to be), *haben* (to have), and *sprechen* (to speak). A1 Connectors, Basic writing.

Building Vocabulary and Conversations: Topics: The Family (*die Familie*), Hobbies (*die Hobbys*), Food and Drinks (*Essen und Trinken*), and Professions (*Berufe*). Grammar: The Accusative case for direct objects. Negation with *nicht* and *kein*. Possessive Articles (*Possessivartikel*): my, your, his, her, etc. (e.g., *mein Buch, deine Tasche*).

Expanding Skills: Talking about the Past: Introduction to the Present Perfect tense (*Perfekt*) to talk about past events (e.g., “I have learned German”). Separable Verbs (*Trennbare Verben*): Understanding verbs like *anrufen* (to call) or *einkaufen* (to shop). Imperative sentence formation.

Textbooks/References

1. *Netzwerk neu A1* (Kursbuch und Arbeitsbuch) by Klett Sprachen.
2. *Netzwerk neu A1* (Vocabulary) by Klett Sprachen.
3. *Grammatik aktiv A1-B1* by Cornelsen.

SEMESTER II

IMA121 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To Study the basic topological properties of the real numbers
- Have the knowledge of the sequence of real numbers and convergence.
- Studying the notion of continuous functions and their properties.
- To gain an understanding of the linear system of equations
- To get introduced to the fundamental concepts of vector spaces
- To impart the basics of linear transformation, orthogonalization, basis, dimensions and eigenvalues.
- To provide the knowledge to apply the concepts of linear algebra in engineering applications

Course Outcomes

- Have a good knowledge of the mathematical concepts in real analysis
- Be able to prove statements and to formulate precise mathematical arguments.
- To solve the problems related to linear systems and matrices
- To apply the knowledge of linear transformation, orthogonal projections, orthonormalization and Least-square solutions in engineering applications.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Reimann Integration, fundamental theorem of calculus.

Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, tests of convergence;

Function of several variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear vector space and examples; linear independence of vectors, basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, norms, orthogonal basis, Gram-Schmidt orthogonalization process, Eigen vectors of a matrix and matrix diagonalization, Applications.

Textbooks/References

1. G. Bartle and D. R. Sherbert, Introduction to Real Analysis, Fourth Edition, Wiley, 2011.

2. T. M. Apostol, Calculus, Volume I, Second Edition, Wiley, 2007.
3. Gilbert Strang, Linear Algebra and Its Applications, 5 edition, Wellesley-Cambridge Press/Siam, 2016
4. K. Hoffman and R. Kunze, Linear Algebra, 2 edition, PHI, 2009
5. Erwin Kreyzig, Advanced Engineering Mathematics, Tenth Edition, Wiley, 2015.

IEC121 Digital Design and Electric Circuits [3-1-2-5]

Course Objectives

The primary objective of this course is to provide the student with the fundamental concepts and skills necessary to analyze and design combinational and sequential logic circuits. The course explains the basics of analog and digital logic circuits. It also introduces the student a hardware description language and its application to the design of combinational, sequential and simple digital systems. The material covered in the lecture is reinforced through practical experience in the associated lab together with the use of Verilog HDL to synthesize logic circuits.

Course Outcomes

- Understand the fundamentals of analog and digital circuits.
- Analyze and design a circuit of logic gates that have the desired relation between the input and output terminals.
- Understand the logic properties of flip flops.
- Analyze and design counters, registers, and similar circuits.
- Implement combinational and sequential circuits using a hardware description language.

Syllabus

Introduction - Analog and Digital circuits – Kirchhoff's Laws, Superposition Theorem, Thevenin's and Norton's Theorems; Review of Number Systems - Number systems and conversions-decimal, binary, 1's and 2's complements, hexadecimal, octal etc. Logic gates-NOT, AND, OR, XOR, XNOR, Universal gates, timing diagrams.

Boolean algebra-DeMorgans theorems, SOP and POS forms. Karnaugh Maps-to simplify Boolean expressions, truth table functions. Combinational Logic-Analyze basic combinational logic circuits, design a combinational logic circuits for a given truth table. Functions of Combinational logic-comparators, adders, code converters, multiplexers, de-multiplexers.

Sequential Circuit Design - Flip-Flops and Latches. SR, D, and JK Flip-Flops. Edge-triggered and Master-Slave Flip-Flops, Excitation table. Counters – Design of asynchronous and synchronous counters. Timing diagrams up/down counters. Shift Registers – data movements in shift registers. SISO, SIPO, PISO, PIPO shift registers.

Memory and programmable logic – RAM, Memory decoding, ROM, PLA, PAL, sequential programmable devices, overview

of logic design using Verilog HDL, Basic concepts, Modules, Ports.

Lab Practice

- Familiarization of Logic Gates.
- Design of Combinational Logic Circuits - Comparators, Adders, Code Converters, Multiplexers, Demultiplexers etc.
- Familiarization of Flip-Flops and Latches. SR, D, and JK Flip-Flops. Edge triggered and Master-Slave Flip-Flops.
- Design of Sequential Logic Circuits, Design of Counters, Asynchronous Counters, Synchronous counters. Shift Registers.
- Simple Verilog HDL programs.

Textbooks/References

1. Floyd, Digital Fundamentals, McGraw Hill, Tenth Edition, 2011.
2. Morris Mano, Digital Circuits and Logic Design", PHI Publication, Fifth Edition, 2015.

ICS121 Data Structures I [3-1-2-5]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees and graphs
- Design and specify algorithms for searching and sorting, and those associated with the above data structures
- Analyse simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences, asymptotic analysis and probabilistic analysis
- Analyse application problems and abstract them to formulate solutions involving data structures and algorithms

Course Outcomes

- Students learn to define operations of data structures like arrays, linked lists, trees and graphs
- Students learn to design and specify algorithms involving above types of data structures
- Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis and probabilistic analysis
- Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms.

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - List- Sorted Lists, Double Linked Lists, Skip list.

Stack & Queue application - Celebrity problem, histogram rectangular area problem.

Binary Trees – Insertion and Deletion of nodes, Tree Traversals,

Polish Notations, Red Black Trees, B-Trees, Heaps, Priority Queues.

Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting – Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching.

Hashing- Application problems on hashing.

Graphs- Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Textbooks/References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005

ICS122 Computer Organization [3-1-0-4]

Course Objectives

- To understand the basics of computer hardware and how software interacts with computer hardware.
- To analyze and evaluate the performance of computers.
- Understand basics of Instruction Set Architecture (ISA) - RISC.
- To understand how computers represent and manipulate data.
- To understand how computer perform arithmetic operations, how they are optimized and made to run faster.
- To understand how the memory management takes place in a computer system.
- To understand what is pipelining, and the design concepts involved.
- Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic unit, control unit, and data, instruction and address flow.

Course Outcomes

- This course will introduce to students the fundamental concepts underlying modern computer organization and architecture.

- Students should be able to know the overall working of a computer.
- Students should be able to get a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed and how programs manipulate and operate on data.
- They should also be able to appreciate how the memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining

Syllabus

Computer abstraction and technology: Basic principles, hardware components, Measuring performance: evaluating, comparing and summarizing performance. Instructions: operations and operands of the computer hardware, representing instructions, making decision, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multi-cycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors.

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies. Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors.

Textbooks/References

1. D. A. Patterson and J. L. Hennessy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003
3. J. L. Hennessy and D. A. Patterson, Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, Zvonko Vranesic and Safwat Zaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

IEC122 Signals and Systems [3-1-0-4]

Course Objectives

- Understand the mathematical description of continuous and discrete time signals and systems.
- Analyze the signals in time domain using convolution sum and Integral.

- Classify signals into different categories based on their properties.
- Analyze Linear Time Invariant (LTI) systems in time and transform domains.

Course Outcomes

- Analyze the different types of signals and systems.
- Determine the linearity, causality, time-invariance and stability properties of continuous and discrete time systems.
- Represent continuous and discrete systems in time and frequency domain using different transforms
- Test whether the system is stable.

Syllabus

Introduction and Classification of signals: Definition of signal and systems, communication and control system as examples Classification of signals. Basic Operations on signals: Amplitude scaling, addition, multiplication, differentiation, integration, time scaling, time shift and time reversal. Elementary signals/Functions: Exponential, sinusoidal, step, impulse and ramp functions.

Expression of triangular, rectangular and other waveforms in terms of elementary signals.

System Classification and properties: Linear, nonlinear, Time variant-invariant, causal, noncausal, static-dynamic, stable-unstable, invertible. Time domain representation of LTI System: Impulse response, convolution sum, convolution integral. Computation of convolution sum and convolution integral using graphical method for unit step and unit step, unit step and exponential, exponential and exponential, unit step and rectangular, and rectangular and rectangular.

LTI system Properties in terms of impulse response: System interconnection, Memory-less, Causal, Stable, Invertible and Deconvolution, and step response.

Fourier Representation of Periodic Signals: CTFS properties and basic problems. Fourier Representation of aperiodic Signals: Introduction to Fourier Transform & DTFT, Definition and basic problems.

Properties of Fourier Transform: Linearity, Time shift, Frequency shift, Scaling, Differentiation and Integration, Convolution and Modulation, Parseval's theorem and problems on properties of Fourier Transform.

The Z-Transforms: Z transform, properties of the region of convergence, properties of the Z-transform, Inverse Z-transform, Causality and stability, Transform analysis of LTI systems.

LAPLACE TRANSFORMS: Introduction, Concept of region of convergence (ROC) for Laplace transforms, constraints on ROC for various classes of signals, Relation between L.T and F.T. of a signal. Laplace transforms of certain signals.

Textbooks/ References

1. Simon Haykins and Barry Van Veen, "Signals and Systems", 2nd Edition, 2008, Wiley India. ISBN 9971-51-239-4.
2. Michael Roberts, "Fundamentals of Signals & Systems", 2nd edition, Tata McGraw-Hill, 2010, ISBN 978-0-07-070221-9.

3. Alan V Oppenheim, Alan S, Willsky and A Hamid Nawab, "Signals and Systems" Pearson Education Asia / PHI, 2nd edition, 1997. Indian Reprint 2002.
4. H.P Hsu, R. Ranjan, "Signals and Systems", Scham's out-lines, TMH, 2006.
5. B. P. Lathi, "Linear Systems and Signals", Oxford University Press, 2005.
4. Rao, K.R., & Paranjpe, A.C.(2016).Psychology in the Indian Tradition. NewDelhi:Springer.
5. Frankl, V.E.(1992). Man's Search for Meaning. Massachusetts:Beacon Press
6. Simanowitz, V., & Pearce, P. (2003). Personality Development. England: Open University Press.

IHS121 Personality Development [1-0-0-1]

Course Objectives

- To understand the basic perspectives of human personality such as; Trait approach, Psychoanalytic approach, Biological basis, Humanistic/phenomenological approach, Behaviorist/learning, Cognitive approach, Interaction perspective, and Transpersonal perspective (Indian and Yoga Psychology).
- Learn to objectively assess and explain the behavior of other people, identify personality traits so as predict how a person will behave, and to help to function effectively.
- Have understanding how hiring decisions are taken based on personality characteristics that serve as requirements of a job.
- Understanding the application of assessment of Type A & B personality on personal health & achievement
- Understand Personality disorders and its identification. Cognitive Behavior Therapy in the context of Psychotherapy for personality disorders.

Course Outcomes

- Understand one's own personality and that of others, appreciate uniqueness of individuals, adapt to people and situations effectively, assess self and others using scientific tools of personality, cope with challenges in life with better understanding of human behavior science.

Syllabus

Personality: Meaning & Assessment. Psychoanalytic & Neo-Psychoanalytic Approach ; Behavioural Approach; Cognitive Approach; Social- Cognitive Approach; Humanistic Approach; The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality etc. Personality disorders; Psychotherapeutic techniques and Yoga & Meditation; Indian perspective on personality; Personality in Sociocultural context.

Textbooks/References

1. Schultz, D.P., & Schultz, S. E. (2005)(8th Edn.)Theories of Personality. Belmont: Thomson Wadsworth.
2. Lindzey, G., Campbell, J.B., & Hall, C.S.(2007)(4th Edn.). Theories of Personality. NewYork:Wiley & Sons
3. Ryckman, R.M. (2008)(9th Edn.).Theories of Personality.Belmont: Thomson Wadsworth.

SEMESTER III

IMA211 Probability, Statistics, and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.
- To introduce various discrete and continuous distributions and concept of estimation theory, confidence interval.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness of fit tests
- To introduce the concept of random processes, Markov chains, Brownian Motion

Course Outcomes

- Define and apply the concepts of probability and conditional probability
- Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions
- Understand the concept and need of hypothesis testing
- Perform the tests for means and variances and Goodness of fit test
- Understand the concept of random processes, Markov chains, Brownian motions

Syllabus

Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulae.

Random variables and distributions: Univariate, Bivariate and multivariate random variables, Cumulative and marginal distribution function, Conditional and multivariate distributions, Functions of random variables: Sum, product, ratio, change of variables.

Mathematical expectations, moments, moment generating function, characteristic functions; Discrete/continuous distributions and limit theorems: Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central limit theorem, Chebyshev's inequality, Law of large numbers.

Estimation Theory: Bias of estimates, Confidence intervals, Minimum variance unbiased estimation, Bayes' estimators, Moment estimators, Maximum likelihood estimators, Chi-square distribution, Confidence intervals for parameters of normal distribution.

Hypothesis testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, Power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Definition and classification of random processes, discrete-time Markov chains, Poisson process, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Textbooks/ References

1. S. Ross, Introduction to Probability and Statistics for and Engineers and Scientists, Third Edition, Elsevier, 2004.
2. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2000.
3. S. M. Ross, Introductory Statistics, Second Edition, Academic Press, 2009.
4. J. Medhi, Stochastic Processes, Third Edition, New Age International, 2009.
5. V.K.Rohati and A.K. Saleh, An introduction to Probability and Statistics, Third Edition. Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, Probability and Random Processes, Oxford University Press, 2001.
7. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, Third Edition., Wiley, 1968.
8. S.M. Ross, Stochastic Processes, Second Edition. Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, Introduction to Probability, Second Edition, Universities Press India, 2009.
10. S.Ross, A First Course in Probability, 10th Edition, Pearson Education, Delhi, 2018

IEC211 Solid State Devices[3-0-0-3]

Course Objectives

- Understanding the basics of semiconductor physics (energy bands, charge carriers, doping, conductivity and mobility)
- To provide students with a sound understanding of existing devices, so that their studies of electronic circuits and systems will be meaningful
- To develop the basic tools with which students can later learn about newly developed devices and applications.
- To develop a deep understanding of the physical structure and operation of MOSFETs, which form the basic building blocks of contemporary very-large-scale-integrated (VLSI) digital circuits such as those for memory and microprocessors.

Course Outcomes

- Develop a basic understanding on the following key concepts in quantum and statistical mechanics relevant to physical properties of electronic materials and their device applications
- Become proficient with the fundamental device physics concepts
- Analyze device characteristics in detail and brainstorm ways towards improving them or adapting them to new applications

Syllabus

Energy Bands and Charge Carriers in Semiconductors: Bonding Forces in Solids, Energy Bands, Direct and Indirect Semiconductors, Variation of Energy Bands with Alloy Composition, Charge Carriers in Semiconductors: Electrons and Holes, Effective Mass, Intrinsic Material, Extrinsic Material, Electrons and Holes in Quantum Wells, Carrier Concentrations: The Fermi Level, Electron and Hole Concentrations at Equilibrium, Temperature Dependence of Carrier Concentrations, Compensation and Space Charge Neutrality, Drift of Carriers in Electric and Magnetic Fields: Conductivity and Mobility, Drift and Resistance, Effects of Temperature and Doping on Mobility, High-Field Effects, The Hall Effect, Invariance of the Fermi Level at Equilibrium, Excess Carriers in Semiconductors: Optical absorption, Luminescence, Carrier lifetime and Photoconductivity, Diffusion of carriers.

Junctions: Fabrication of p-n Junctions, Equilibrium Conditions, Forward- and Reverse-Biased Junctions; Steady state conditions, Reverse-Bias Breakdown, Transient and A-C Conditions, Metal-Semiconductor Junctions: Schottky Barriers, Rectifying Contacts, Ohmic Contacts, Typical Schottky Barriers.

Field-Effect Transistors: Transistor Operation, The Junction FET, The Metal-Semiconductor FET, The GaAs MESFET, The High Electron Mobility Transistor (HEMT), Short Channel Effects, The Metal-Insulator-Semiconductor FET, The Ideal MOS Capacitor, Current-Voltage Characteristics of MOS Gate Oxides, The MOS Field-Effect Transistor, Mobility Models, Short Channel MOSFET I-V Characteristics, Control of Threshold Voltage, Substrate Bias Effects—the “body” effect, Subthreshold Characteristics, Equivalent Circuit for the MOSFET, MOSFET Scaling and Hot Electron Effects, Drain-Induced Barrier Lowering, Short Channel Effect and Narrow Width Effect, Gate-Induced Drain Leakage.

Bipolar Junction Transistors: Fundamentals of BJT Operation, Minority carrier distributions and terminal currents, Generalized biasing, Switching, Drift in the Base Region, Base Narrowing, Avalanche Breakdown, Injection Level; Thermal Effects, Base Resistance and Emitter Crowding, Kirk Effect, Frequency limitations of Transistors.

Textbooks/References

1. Solid State Electronic Devices, Ben G. Streetman and Sanjay Banerjee, 7th Edition, Pearson Education India
2. Semiconductor physics and devices, Donald A. Neamen, 4th Edition, McGraw Hill Education
3. Physics of Semiconductor Devices, S.M. Sze, Kwok K. Ng, Third Edition, John Wiley & Sons, Inc.
4. Semiconductor Device Fundamentals, Robert F. Pierret, Pearson Education India

IEC212 Analog Electronics [3-0-2-4]

Course Objectives

- Design and analyze various BJT and FET amplifier configurations

- Construct frequency response of FET amplifiers at various frequencies.
- Analyze various feedback amplifiers
- Analyze Power amplifier circuits in different modes of operation.
- Construct various oscillator.

Course Outcomes

- Understand the small signal analysis of BJTs and FET
- Design and analyze BJT and FET amplifier circuits.
- Design sinusoidal and non-sinusoidal oscillators.
- Design power amplifier circuits

Syllabus

Small signal operation and Models: Collector current and transconductance, Base current and input resistance, Emitter current and input resistance, voltage gain, Separating the signal and the DC quantities, The hybrid II model. AC analysis for CE fixed bias, voltage divider bias, emitter bias amplifier under fixed bias, voltage divider bias with and without bypassed configurations, Emitter follower configuration.

Small signal operation and modeling: The DC bias point, signal current in drain, voltage gain, small signal equivalent circuit models, transconductance. MOSFET Amplifier configuration: Basic configurations, characterizing amplifiers, CS amplifier with and without source resistance R_s , Source follower. MOSFET internal capacitances and High frequency model: The gate capacitive effect, Junction capacitances, High frequency model. Frequency response of the Cs amplifier: The three frequency bands, high frequency response, Low frequency response.

Current mirror circuit, Multistage Amplifier, Darlington Amplifier – Bootstrap technique – Cascade, Cascode configurations – Differential amplifier.

Feedback Amplifier: General feedback structure, Properties of negative feedback, The Four Basic Feedback Topologies, The series-shunt, series-series, shunt-shunt and shunt-series amplifiers (Qualitative Analysis).

Oscillators: Barkhausen Criteria, Oscillator operation, FET based RC Phase shift oscillator, Wein bridge oscillator, LC Oscillators: Hartley Oscillator, Colpitts Oscillator and Crystal Oscillators

Output Stages and Power Amplifiers: Introduction, Classification of output stages, Class A output stage, Class B output stage: Transfer Characteristics, Power Dissipation, Power Conversion efficiency, Class AB output stage, Class C tuned Amplifier.

Lab Practice

Design and setup the Common Source JFET/MOSFET amplifier and plot the frequency response.

Design and set up the BJT common emitter voltage amplifier with and without feedback and determine the gain- bandwidth product, input and output impedances.

Design Adder, Integrator and Differentiator circuits using Op-Amp Test a comparator circuit and design a Schmitt trigger for the given UTP and LTP values and obtain the hysteresis Design Monostable and a stable Multivibrator using 555 Timer.

Textbooks/References

1. Microelectronic Circuits, Theory and Applications, Adel S Sedra, Kenneth C Smith, 6th Edition, Oxford, 2015.ISBN:978-0-19808913-1
2. Op-Amps and Linear Integrated Circuits, Ramakant A Gayakwad, 4 th Edition. Pearson Education, 2000. ISBN: 8120320581
3. Electronic Devices and Circuit Theory, Robert L Boylestad and Louis Nashelsky, 11 th Edition, Pearson Education, 2013, ISBN: 978- 93-3254260-0.
4. Fundamentals of Microelectronics, BehzadRazavi, 2 nd Edition, John Wiley, 2015, ISBN 978-81-265-7135-2.
5. J.Millman& C.C.Halkias-Integrated Electronics, 2 nd edition, 2010, TMH. ISBN 0- 07462245-5.

IEC213 Microprocessors and Microcontrollers [3-0-2-4]

Course Objectives

- Familiarize the importance and applications of microprocessors and microcontrollers
- Expose architecture of 8086 processor and 8051 microcontroller
- Impart design and coding knowledge on 8086 and 8051 family.
- Understand the interrupt system of 8086 and 8051 and the use of interrupts
- Understand the operation and use of inbuilt Timers / Counters and Serial port of 8051.
- Interface 8086 and 8051 to external memory and I/O devices using its I/O ports.

Course Outcomes

- Design and develop assembly language/C code to solve problems using 8086/8051
- Gain the knowledge for interfacing various devices to 8086 family and 8051
- Demonstrate design of interrupt routines for interfacing devices
- Know functioning of hardware devices and interfacing them to x86 and 8051

Syllabus

8086 architecture, addressing modes, machine language instruction formats, instruction set, programming using assembly language and C, memory interfacing, interrupts, interrupt programming, macros, 8086 bus configuration and timings, physical memory organization, I/O addressing capability, minimum mode, maximum mode operations, basic I/O interfacing. 8086 microprocessors interfacing: control signal generation, interfacing: Memory-mapped-IO and IO-mapped-IO, interfacing

memory, interfacing with 8255, ADC and DAC circuits, memory banking, memory segmentation, implementing loops, delays and counters, Subroutines, Multiple interrupts & priority, DMA, serial IO lines.

8051 Architecture- Registers, Pin diagram, I/O ports functions, Internal Memory organization. External Memory (ROM & RAM) interfacing. 8051 Instruction Set: Addressing Modes, Data Transfer instructions, Arithmetic instructions, Logical instructions, Branch instructions, Bit manipulation instructions. Assembly language program examples.

8051 Stack, I/O Port Interfacing and Programming: 8051 Stack, Stack and Subroutine instructions. Assembly language program examples on subroutine

8051 Timers and Serial Port: 8051 Timers and Counters – Operation and Assembly language programming to generate a pulse using Mode-1 and a square wave using Mode- 2 on a port pin. 8051 Serial Communication- Basics of Serial Data Communication, RS-232 standard, 9 pin RS232 signals, Simple Serial Port programming in Assembly and C to transmit a message and to receive data serially.

8051 Interrupts and Interfacing Applications: 8051 Interrupts. 8051 Assembly language programming to generate an external interrupt using a switch, 8051 C programming to generate a square waveform on a port pin using a Timer interrupt. Interfacing 8051 to ADC-0804, DAC, and Stepper motor.

Lab Practice

Data Transfer: Block Move, Exchange, Sorting, Finding largest element in an array. Arithmetic Instructions: Addition, subtraction, multiplication and division, square, Cube (16 bits Arithmetic operations – bit addressable). Counters.

Boolean & Logical Instructions (Bit manipulations). conditional CALL and RETURN. Code conversion: BCD – ASCII; ASCII – Decimal; Decimal - ASCII; HEX - Decimal and Decimal -HEX. Programs to generate delay, Programs using serial port and on-Chip timer/counter.

Interface a simple toggle switch to 8051 and write an ALP to generate an interrupt which switches on an LED (i) continuously as long as switch is on and (ii) only once for a small time when the switch is turned on. Write a program to (i) transmit and (ii) to receive a set of characters serially by interfacing 8051 to a terminal. Write ALPs to generate waveforms using ADC interface. Write ALP to interface a Stepper Motor to 8051 to rotate the motor.

Textbooks/References

1. Douglas Hall, SSSP Rao "Microprocessor and Interfacing", 3rd edition, Tata Mc Graw Hill, 2012.
2. Kenneth J Ayala, "The 8086 Microprocessor programming and Interfacing the PC" Cengage Learning, 2011.
3. Lyla B das, "Microprocessors and Microcontrollers", Pearson education India, 2nd edition, 2014
4. Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D. McKinlay, The 8051 Microcontroller and Embedded Systems using assembly and C", PHI, 2006 / Pearson, 2006.
5. Kenneth J. Ayala, The 8051 Microcontroller, 3rd Edition, Thomson/Cengage Learning.

6. The 8051 Microcontroller Based Embedded Systems”, Manish K Patel, McGraw Hill, 2014, ISBN: 978-93-329-0125-4

IEC214 Principles of Communication [3-1-0-4]

Course Objectives

This course will enable students to

1. Understand and analyse concepts of Analog Modulation schemes viz; AM, FM., Low pass sampling and Quantization as a random process.
2. Understand and analyse concepts digitization of signals viz; sampling, quantizing and encoding.
3. Evolve the concept of SNR in the presence of channel induced noise and study Demodulation of analog modulated signals.
4. Evolve the concept of quantization noise for sampled and encoded signals and study the concepts of reconstruction from these samples at a receiver.

Course Outcomes

After studying this course the students will be able to

- Analyze and compute performance of AM and FM modulation in the presence of noise at the receiver.
- Analyze and compute performance of digital formatting processes with quantization noise.
- Multiplex digitally formatted signals at Transmitter and demultiplex the signals and reconstruct digitally formatted signals at the receiver.
- Design/Demonstrate the use of digital formatting in Multiplexers, Vocoders and Video transmission.

Syllabus

AMPLITUDE MODULATION: Introduction, Amplitude Modulation: Time & Frequency Domain description, Switching modulator, Envelop detector.

DOUBLE SIDE BANDSUPPRESSED CARRIER MODULATION: Time and Frequency Domain description, Ring modulator, Coherent detection, Costas Receiver, Quadrature Carrier Multiplexing.

SINGLE SIDE-BAND AND VESTIGIAL SIDEBAND METHODS OF MODULATION: SSB Modulation, VSB Modulation, Frequency Translation, Frequency- Division Multiplexing, Theme Example: VSB Transmission of Analog and Digital Television.

ANGLE MODULATION: Basic definitions, Frequency Modulation: Narrow Band FM, Wide Band FM, Transmission bandwidth of FM Signals, Generation of FM Signals, Demodulation of FM Signals, FM Stereo Multiplexing, Phase-Locked Loop: Nonlinear model of PLL, Linear model of PLL, Nonlinear Effects in FM Systems. The Superhetrodyne Receiver.

NOISE - Shot Noise, Thermal noise, White Noise, Noise Equivalent Bandwidth, NOISE IN ANALOG MODULATION: Introduction, Receiver Model, Noise in DSB-SC receivers. Noise in AM receivers, Threshold effect, Noise in FM receivers, Capture effect, FM threshold effect, FM threshold reduction, Pre-emphasis and De-emphasis in FM

SAMPLING AND QUANTIZATION: Introduction, Why Digitize Analog Sources?, The Low-pass Sampling process Pulse Amplitude Modulation. Time Division Multiplexing, PulsePosition Modulation, Generation of PPM Waves, Detection of PPM Waves SAMPLING AND QUANTIZATION (Contd): The Quantization Random Process, Quantization Noise, Pulse-Code Modulation: Sampling, Quantization, Encoding, Regeneration, Decoding, Filtering, Multiplexing; Differential PCM, Delta Modulation, Adaptive delta modulation, Application examples - (a) Video + MPEG and (b) Vocoders.

Textbooks/References

1. Simon Haykins & Moher , Communication Systems, 5th Edition, John Wiley, India Pvt. Ltd, 2010, ISBN 978 – 81 – 265 – 2151 – 7.
2. B. P. Lathi, Modern Digital and Analog Communication Systems, Oxford University Press., 4th edition.
3. Simon Haykins, An Introduction to Analog and Digital Communication, John Wiley India Pvt. Ltd.,2008, ISBN 978-81-265-3653- 5.
4. H.Taub & D.L.Schilling, Principles of Communication Systems, TMH,2011
5. Harold P.E, Stern Samy and A.Mahmond, Communication Systems, Pearson Edition, 2004.

IEC215 Linear Integrated Circuits [3-1-0-4]

Course Objectives

- Understand the basic operational amplifier principle and working.
- To develop the skill of analysis and design of various circuits using operational amplifiers.
- To develop design skills to design various circuits using different data conversion systems.
- To develop the skill of analysis and design of various circuits using 555 timers.

Course Outcomes

- Associate and apply the concepts of operational amplifiers to design
- Operational amplifier circuits for various applications
- Design Oscillators and active filters using op-amps.
- Test and validate symbol processing and performance parameters at the receiver under ideal and corrupted bandlimited channels.
- Explain the working and applications of timer, VCO and PLL ICs.

Syllabus

Review of fundamentals: Amplifiers, Need for amplification, signal amplification, voltage gain, current gain, power gain, Gain in dB, Circuit Model of typical voltage amplifier, effect of loading, linearity, distortion, frequency response, bandwidth.

Differential amplifier: Basic structure and principle of operation, calculation of differential gain, common mode gain, CMRR and ICMR, introduction to operational amplifiers: concept of negative feedback and virtual short, difference amplifier, ideal operational amplifier, op-amp terminals, function and characteristics of ideal op amp, differential and common-mode signals, inverting and noninverting configurations, closed-loop gain, effect of finite open-loop gain, input and output resistances, weighted summer, voltage follower, difference amplifiers, instrumentation amplifier, integrators and differentiators, inverting configuration with general impedances.

Log and antilog amplifiers, analog multipliers, voltage comparators-Schmitt trigger, square wave generator, monostable multivibrator, triangular wave generator, oscillators-RC phase shift oscillator, Wein Bridge oscillator. Linear halfwave rectifiers, precision rectifiers, peak detectors, DC performance: Input bias current, input offset current, input offset voltage, AC performance: bandwidth, slew rate. Active filters: Butterworth first order and second order function for LPF, HPF, BPF and notch filters.

Data converters-definitions and specifications – Digital-to-analog converters (DAC) : weighted resistor and R-2R DAC, resistor string, Analog-to-digital converters (ADC) : Single slope, dual slope, successive approximation, flash, integrating type, counter ramp, successive approximation, & oversampling ADCs - sigma - delta ADC. Single slope, dual slope, successive approximation, flash etc. Timer IC 555 – internal diagram – working - multivibrators with timer IC 555. Phase-locked-loops – Introduction, basic principles, phase detector, VCO, low-pass filter.

Tutorial

- Simulation of linear circuits like Inverting Amplifier, Non-inverting Amplifier, Difference Amplifier, Summing Amplifier etc. using LTSpice
- Simulation of non-linear circuits like Integrator, Differentiator, Logarithmic Amplifier, Precision Rectifiers etc.
- Analysis of circuits and numerical problem solving
- Simulation experiments for waveform generation and data conversion
- Design of active filters and their simulation using LTSpice
- Mini project based on an application using Linear Integrated Circuits

Textbooks/References

1. Sergio Franco, Design with Operational Amplifiers & Analog integrated Circuits, McGraw Hill.
2. Fiore J.M., Operational Amplifiers and Linear Integrated Circuits, Jaico Publishing House.
3. Coughlin R.F. & Driscoll F.F., Operational Amplifiers and Linear Integrated Circuits, Pearson Education.
4. Op-amps and Linear Integrated Circuits, Ramakant A. Gayakwad, Fourth Edition, Pearson Education.
5. Adel S. Sedra, Kenneth C. Smith, Microelectronic Circuits, Seventh Edition, Oxford University Press.

ICS215 Data Structures II [1-0-2-2]

Course Objectives

- Teach programming with emphasis on problem solving and introduce Data structures.
- Provide the Foundations of the practical implementation and usage of Algorithms and Data Structures.

Course Outcomes

- Design correct programs to solve problems.
- Choose efficient data structures and apply them to solve problems.

Syllabus

Problems on Geometric algorithms but not limited to: Klee's Algorithm, Manhattan distance problems, Collinear checking, Identifying Integral points inside a Triangle, Circumcenter of a Triangle, Triangular Matchstick Number, area of Circumcircle of an Equilateral Triangle, Number of rectangles in $N * M$ grid, Area of two overlapping rectangles, Number of unique rectangles formed using N unit squares, Circle and Lattice Points, Pizza cut problem.

Algorithms on Bit Manipulation but not limited to: Letter manipulation problems, k-th bit Manipulation problems, Kernighan's Algorithm to count set bits in an integer.

Discussion on Numerical algorithms but not limited to: Gauss-Jordan Elimination, Matrix Manipulation problems.

Textbooks/References

1. Cormen, Thomas H., et al. Introduction to algorithms. MIT press, 2009.
2. Aho, Albrecht V., J. E. Hopcroft, and J. D. Ullman. "Data structures and algorithms (1983).
3. Drozdek, Adam. Data Structures and algorithms in C^{++} . Cengage Learning, 2012.
4. Allen, Weiss Mark. Data structures and algorithm analysis in C^{++} . Pearson Education India, 2007.
5. Kleinberg, Jon, and Eva Tardos. Algorithm design. Pearson Education India, 2006.
6. Skiena, Steven S. The algorithm design manual. Springer International Publishing, 2020.
7. Knuth, Donald Ervin. The art of computer programming. Vol. 3. Pearson Education, 1997.
8. Nomura, Seiichi. "C Programming and Numerical Analysis: An Introduction." Synthesis Lectures on Mechanical Engineering 2.2 (2018): 1-198.
9. Dasgupta, Sanjoy, C. H. Papadimitriou, and U. V. Vazirani. "Algorithms; 2006."
10. Trefethen, Lloyd N., and David Bau III. Numerical linear algebra. Vol. 50. Siam, 1997.

SEMESTER IV

IMA221 Differential Equations and Transforms [3-1-0-4]

Course Objectives

- Find the Fourier series representation of a function of one variable
- Introduce the Fourier series and its application to the solution of partial differential equations
- Introduce the concepts of Laplace and Fourier transforms.
- Identify the type of a given differential equation and select and apply the appropriate analytical technique for finding the solution of first order and selected higher order ordinary differential equations.
- Introduce students to partial differential equations
- Introduce students to how to solve linear Partial Differential with different methods.

Course Outcomes

- Analyse and solve engineering problems using Fourier series.
- Find the Laplace and Fourier transforms of functions of one variable.
- Solve first order differential equations utilizing the standard techniques for separable, exact, linear, homogeneous, or Bernoulli cases. Find particular solutions when given initial or boundary conditions.
- Will be able to find solution of higher-order linear differential equations
- Classify PDEs, apply analytical methods, and physically interpret the solutions.

Syllabus

Fourier Series: Dirichlet's conditions – General Fourier series – Odd and even functions – Half range Sine and Cosine series – Complex form of Fourier series – Parseval's identity – Harmonic Analysis. Convergence of FS, differentiation and integration of Fourier series.

Laplace and Fourier Transforms: Laplace transform pair, Fourier Integral Theorem – Fourier transform pair - Sine and cosine transforms – Properties – Transform of elementary functions – Convolution theorem – Parseval's identity.

Ordinary Differential Equations: Method of variation of parameters – Laplace transform method– Homogenous equation of Euler's and Legendre's type – System of simultaneous linear differential equations with constant coefficients.

Partial Differential Equations Formation – Solutions of first order equations – Standard types and Equations reducible to standard types – Singular solutions – Lagrange's linear equation – Integral surface passing through a given curve – Classification of partial differential equations - Solution of linear equations of higher order with constant coefficients – Linear non-homogeneous

Fourier Series Solutions of Partial Differential Equations:

Method of separation of variables – Solutions of one dimensional wave equation and one dimensional heat equation – Steady state solution of two-dimensional heat equation – Fourier series solutions in Cartesian coordinates.

Textbooks/References

1. C. Edwards and D. Penney, Elementary Differential Equations with Boundary Value Problems, 6th edition, Pearson, 2003
2. W.E. Boyce and R.C. DiPrima, Elementary Differential Equations, 7th Ed., John Wiley & Sons, 2002.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley, 2015.
4. Tyn Myint-U, L. Debnath, Linear Partial Differential Equations for Scientists and Engineers, 4th Edition, Birkhauser, 2007.

IEC221 Digital Design with Hardware Description Languages [3-0-2-4]

Course Objectives

This course will enable students to

- Design digital systems in a larger system design context.
- Understand the concepts of Verilog HDL
- Implementation of digital systems using Verilog HDL

Course Outcomes

After studying this course, students will be able to:

- Construct the combinational circuits, using discrete gates
- Design a sequential circuit
- Describe Verilog model for combinational and sequential circuits and test pattern generation.

Syllabus

Revision of basic Digital systems: Combinational Circuits, Sequential Circuits, Timing, set-up and hold time, Power Dissipation, Mealy and Moore Machines, Equivalent States and Reduction of State Tables

Current state of the field: SoC, IP Design, Design Methodology, System modeling, Hardware Software Co-design, Application Domains.

Digital System Design: Top-Down approach to design, Datapath, Control-path, Pipelining, Resource sharing Introduction to HDL: (VHDL/Verilog): (Use either VHDL or Verilog for RTL).

Computer Aided Design, Hardware Description Languages, Verilog description of combinational circuits, Verilog modules and assignments, Modeling flip flops using an Always Block, Delays in Verilog, Compilation, Simulation and Synthesis of Verilog code, Verilog datatypes and operators, Verilog models for multiplexers, Modeling Registers using Always statements, Behavioral and Structural Verilog.

Introduction to Programmable Logic Devices: Overview of Programmable Logic Devices, Simple Programmable Logic Devices (SPLDs), Programmable Logic Arrays (PLAs), Programmable

Logic Devices (PLDs), Complex Programmable Logic Devices (CPLDs), Field Programmable Gate Arrays (FPGAs), Design flow for FPGAs

Lab Practice

Implementation of combinational and sequential designs using Verilog HDL.

Textbooks/ References

1. Charles Roth, Lizy K John, Byeong Kil Lee, Digital System Design Using Verilog
2. Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design
3. Peter J. Ashenden, - Digital Design: An Embedded Systems Approach Using VERILOG, Elsevier, 2010.

IEC222 Digital Communication [3-0-2-4]

Course Objectives

- Understand the mathematical representation of signal, symbol, and noise.
- Understand the concept of signal processing of digital data and signal conversion to symbols at the transmitter and receiver.
- Compute performance metrics and parameters for symbol processing and recovery in ideal and corrupted channel conditions.
- Compute performance parameters and mitigate channel induced impediments in corrupted channel conditions.

Course Outcomes

- Associate and apply the concepts of Bandpass sampling to well specified signals and channels.
- Analyze and compute performance parameters and transfer rates for low pass and bandpass symbol under ideal and corrupted non band limited channels.
- Test and validate symbol processing and performance parameters at the receiver under ideal and corrupted band-limited channels.
- Demonstrate that bandpass signals subjected to corruption and distortion in a band-limited channel can be processed at the receiver to meet specified performance criteria.

Syllabus

Baseband and pass band signal representations, Bandpass Signal to Equivalent Low pass: Hilbert Transform, Pre-envelopes, Complex envelopes, Canonical representation of bandpass signals, Complex low pass representation of bandpass systems, Complex representation of band pass signals and systems.

Line codes: unipolar NRZ, polar NRZ, unipolar RZ, bipolar RZ, Manchester code, differential encoding, Power spectral densities of lines coding techniques.

Baseband pulses and matched filter detection, probability of

error due to noise, intersymbol interference, Nyquist's criterion for distortionless transmission, pulse shaping, raised cosine spectrum, equalization, eye pattern, signal-space analysis: geometric representation of signals, Gram-Schmidt Orthogonalization procedure, conversion of continuous AWGN channel into a vector channel, maximum-likelihood decoding, correlation receiver, equivalence of correlation and matched filter receivers. Passband data transmission, complex representation of band pass signals and systems, coherent schemes: binary amplitude-shift keying (BASK), frequency shift keying (BFSK), phase shift keying (BPSK), quadriphase-shift keying (QPSK), M-ary PSK, M-ary QAM, generation and detection, bandwidth requirements, calculation of average probability error using MAP/ML algorithms, constellation diagram, differential phase-shift keying. Non-coherent orthogonal modulation techniques: BFSK, differential phase shift keying (DPSK), Block diagrams treatment of Transmitter and Receiver, Probability of error. Spread spectrum modulation, pseudo-noise sequences, generation of PN sequences, properties of maximal length sequences, spread spectrum systems, direct sequence spread spectrum with coherent BPSK, processing gain, jamming margin, effect of despreading on a narrowband interference, some applications of direct sequence spread spectrum signals.

Lab Practice

Hardware experiments: Sampling and reconstruction of low pass signals, time division multiplexing/demultiplexing, generation and detection of BASK, BFSK and BPSK, generation of PN sequence.

Simulation experiments: Pulse Code Modulation (PCM), transmitter, receiver and performance analysis of BPSK, QPSK using MATLAB/Scilab/Python.

Textbooks/References

1. Simon Haykin, Communication Systems, 4th edition, John Wiley Inc.
2. Modern Digital and Analog Communication Systems, B. P. Lathi, Oxford University Press., 4th edition.
3. John G Proakis and Masoud Salehi, "Fundamentals of Communication Systems", 2014 Edition, Pearson Education, ISBN 978-8-131-70573-5
4. Bernard Sklar and Ray, "Digital Communications - Fundamentals and Applications", Pearson Education, Third Edition, 2014, ISBN: 978-81- 317-2092-9.
5. Ian A Glover and Peter M Grant, "Digital Communications", Pearson Education, Third Edition, 2010, ISBN 978-0-273-71830-7

IEC223 Electromagnetic Theory [3-1-0-4]

Course Objectives

- Study the different coordinate systems, Physical significance of Divergence, Curl and Gradient.
- Understand the applications of Coulomb's law and Gauss law to different charge distributions and the applications of Laplace's and Poisson's Equations to solve real time problems on capacitance of different charge distributions.

- Understand the physical significance of BiotSavart's, Amperes's Law and Stoke's theorem for different current distributions.
- Infer the effects of magnetic forces, materials and inductance.
- Know the physical interpretation of Maxwell's equations and applications for Plane waves for their behavior in different media.
- Acquire knowledge of Poynting theorem and its application of power flow.

Course Outcomes

- Evaluate problems on electrostatic force, electric field due to point, linear, volume charges by applying conventional methods and charge in a volume.
- Apply Gauss law to evaluate Electric fields due to different charge distributions and Volume Charge distribution by using Divergence Theorem.
- Determine potential and energy with respect to point charge and capacitance using Laplace equation and Apply Biot-Savart's and Ampere's laws for evaluating Magnetic field for different current configurations
- Calculate magnetic force, potential energy and Magnetization with respect to magnetic materials and voltage induced in electric circuits.
- Apply Maxwell's equations for time varying fields, EM waves in free space and conductors and Evaluate power associated with EM waves using Poynting theorem

Syllabus

Revision of Vector Calculus - Coulomb's Law, Electric Field Intensity and Flux density: Experimental law of Coulomb, Electric field intensity, Field due to continuous volume charge distribution, Field of a line charge, Field due to Sheet of charge, Electric flux density, Numerical Problems.

Gauss's law and Divergence: Gauss's law, Application of Gauss's law to point charge, line charge, Surface charge and volume charge, Point (differential) form of Gauss law, Divergence. Maxwell's First equation (Electrostatics), Vector Operator ∇ and divergence theorem, Numerical Problems.

Energy, Potential and Conductors: Energy expended or work done in moving a point charge in an electric field, The line integral, Definition of potential difference and potential, The potential field of point charge, Potential gradient, Numerical Problems. Current and Current density, Continuity of current.

Poisson's and Laplace's Equations: Derivation of Poisson's and Laplace's Equations, Uniqueness theorem, Examples of the solution of Laplace's equation, Numerical problems on Laplace equation Steady Magnetic Field: Biot-Savart Law, Ampere's circuital law, Curl, Stokes' theorem, Magnetic flux and magnetic flux density, Basic concepts Scalar and Vector Magnetic Potentials, Numerical problems.

Magnetic Forces: Force on a moving charge, differential current elements, Force between differential current elements, Numerical problems. Magnetic Materials: Magnetization and

permeability, Magnetic boundary conditions, The magnetic circuit, Potential energy and forces on magnetic materials, Inductance and mutual reactance, Numerical problems.

Faraday's law of Electromagnetic Induction –Integral form and Point form, Numerical problems.

Maxwell's equations Continuity equation, Inconsistency of Ampere's law with continuity equation, displacement current, Conduction current, Derivation of Maxwell's equations in point form, and integral form, Maxwell's equations for different media, Numerical problems.

Uniform Plane Wave: Plane wave, Uniform plane wave, Derivation of plane wave equations from Maxwell's equations, Solution of wave equation for perfect dielectric, Relation between E and H, Wave propagation in free space, Solution of wave equation for sinusoidal excitation, wave propagation in any conducting media ($\gamma, \alpha, \beta, \eta$) and good conductors, Skin effect or Depth of penetration, Poynting's theorem and wave power, Numerical problems.

Textbooks/References

1. W.H. Hayt and J.A. Buck, Engineering Electromagnetics, 8th Edition, Tata McGraw Hill, 2014, ISBN-978-93-392-0327-6.
2. Elements of Electromagnetics Matthew N.O., Sadiku, Oxford university press, 4th edition
3. Electromagnetic Waves and Radiating systems, E. C. Jordan and K.G. Balmain, PHI, 2nd Edition
4. Electromagnetics Joseph Edminister, Schaum Outline Series, McGraw Hill.

ICS224 Computer Networks [3-0-2-4]

Course Objectives

- The students should understand the layers of networking devices.
- They should be familiar with a few networking protocols.
- They should study the different types of networks and topologies of networks.

Course Outcomes

- To distinguish the importance of different networking components.
- To understand the functionalities of each networking layers and standards.
- To write simple networking based programs at real and simulator level.

Syllabus

Evolution of computer networks: Network Architecture-OSI, TCP/IP models.

Physical and Data link layer: Encoding, Framing, Error detection, HDLC, PPP, sliding window protocols, medium access control, Token Ring, Wireless LAN, Packet Switching.

Network Layer: Internet addressing, IP, ARP, ICMP, CIDR, Routing algorithms (RIP, OSPF, BGP).

Transport Layer: UDP, TCP, flow control, congestion control, Introduction to quality of service.

Application Layer: DNS, Web, HTTP, email, authentication, encryption.

Lab Practice

Unix network measurement and analysis tools, NS3 Socket interface and programming, RPC, RMI, Assignments using Network Simulators.

Textbooks/References

1. L. L. Peterson and B. S. Davie, Computer Networks: A Systems Approach, Fifth Edition, Elsevier, 2011.
2. A. S. Tanenbaum and D. J. Wetherall, Computer Networks, Fifth Edition, Pearson, 2011.
3. W. R. Stevens, UNIX Network Programming, Volume 1: Networking APIs: Sockets and XTI, Second Edition, Prentice Hall, 1998.
4. S. S. Panwar, S. Mao, J. Ryoo, and Y. Li, TCP/IP Essentials: A Lab-based Approach, Cambridge Press, 2004.
5. J. F. Kurose and K. W. Ross, Computer Networking: A Top Down Approach, Seventh Edition, Pearson India, 2017.
6. D. E. Comer, Internetworking with TCP/IP Vol. 1, Sixth Edition, Prentice Hall of India, 2006.
7. B. Forouzan, Data Communications and Networking, Fifth Edition, Tata McGraw Hill, 2012.
8. Introduction to Network Simulator NS2, Second Edition, 2011

ICS225 Data Structures III [1-0-2-2]

Course Objectives

- Ensure that the student evolves into a competent programmer capable of designing and analyzing implementations of algorithms and data structures for different kinds of problems.
- Expose the student to the algorithm analysis techniques.

Course Outcomes

- Analyze the efficiency of programs based on Complexity.
- Understand the necessary mathematical abstraction to solve problems.

Syllabus

Algorithms on Graph connectivity but not limited to: Tarjan's and Kosaraju's strongly connected components algorithms, Detect cycles in an undirected graph, Degree of vertices in a Graph, Path identification between vertices in Undirected graph.

Discussions on Randomized algorithms but not limited to: Reservoir Sampling, Birthday Paradox, Load Balancing problem, Karger's algorithm for Minimum Cut, Freivald's Algorithm to check the product of a matrix, Monte Carlo estimation.

Branch and Bound Algorithms: Knapsack problem, Travelling salesman problem.

Threaded Binary Tree, Splay trees, Foldable binary trees, Additional problems on BST, Binomial heap, Fibonacci heap, Topological sorting, self-organizing tree, segment tree, Binary indexed tree, suffix array and suffix tree, pattern searching, Tribonacci word.

Textbooks/References

1. Cormen, Thomas H., et al. Introduction to algorithms. MIT press, 2009.
2. Aho, Albred V., J. E. Hopcroft, and J. D. Ullman. "Data structures and algorithms (1983).
3. Drozdek, Adam. Data Structures and algorithms in C++. Cengage Learning, 2012.
4. Allen, Weiss Mark. Data structures and algorithm analysis in C + +. Pearson Education India, 2007.
5. Kleinberg, Jon, and Eva Tardos. Algorithm design. Pearson Education India, 2006.
6. Skiena, Steven S. The algorithm design manual. Springer International Publishing, 2020.
7. Knuth, Donald Ervin. The art of computer programming. Vol. 3. Pearson Education, 1997.
8. Nomura, Seiichi. "C Programming and Numerical Analysis: An Introduction." Synthesis Lectures on Mechanical Engineering 2.2 (2018): 1-198.
9. Dasgupta, Sanjoy, C. H. Papadimitriou, and U. V. Vazirani. "Algorithms; 2006."
10. Trefethen, Lloyd N., and David Bau III. Numerical linear algebra. Vol. 50. Siam, 1997.

IHS221 Fundamentals of Economics [1-0-0-1]

Course Objectives

- To familiarize the participants concepts and techniques in Economics
- To make the participants appreciate the applications of core concepts in economics for managerial decision making
- To sensitize the participants how economic environment affects Organizations

Course Outcomes

- It will help the students to analyze the demand and supply conditions and assess the positions of a company.
- It will help to design competition strategies, including costing, pricing, product differentiation and market environment according to the natures of products and structures of market

Syllabus

Introduction to Fundamentals of Economics

Micro & Macro Economics, Managerial Economics – Definition – Nature & Scope, Fundamental concepts in Managerial economics for decision making: Incremental Principle, Opportunity Cost, Discounting Principle, Time Concept, Equi-Marginal Principle – Illustrations, Decision Making – Process and Conditions – Difference between Risk & Uncertainty.

Demand Analysis and Forecasting

Meaning of Demand – Types of Demand – Law of Demand & its Exceptions, Elasticity of Demand – Price Elasticity, Income Elasticity, Cross Elasticity, Promotion Elasticity, Applications of the concepts of Elasticity, Demand Forecasting – Process – Statistical & Non-Statistical Techniques, Utility Analysis & Consumer Behaviour – Equilibrium of the consumer using Cardinal & Ordinal Utility (Indifference Curve) Theories.

Supply & Production

Theory of Production – Meaning of Production function, Production function with one variable input – Law of Variable Proportions – Returns to Scale, Production function with two variable inputs – Iso-quants – Producers' Equilibrium, Economies of Scale – Types – Economies of Scope, Theory of Costs – Classification of Costs - Short Run & Long Run Cost Curves, Revenue Curves.

Market Structure

Market – Meaning & Elements, Classification of Markets – Markets based on Competition, Theory of Firm – Profit Maximization Rules, Price & Output Determination under Perfect Competition, Price & Output Determination under Monopoly – Monopoly Price Discrimination, Price & Output Determination under Monopolistic Competition, Price & Output Determination under Oligopoly – Kinked Demand curve model only.

Macro Economic Concepts

National Income Concepts – Measurement of National Income, An overview of Financial System in India, An overview of Fiscal & Monetary Policies in India, Balance of Payments: Causes of Disequilibrium & Remedies, Inflation in India – Causes & Remedies. Free Market Economy & Need for Government Intervention – An appraisal of Economic Reforms in India.

Textbooks/References

1. Dwivedi D.N, Managerial Economics, Vikas Publications (ISBN 8125910042)
2. P.L. Mehta, Managerial Economics Analysis, Problems and Cases - Sultan Chand & Sons (ISBN 81-7014-386-1)
3. K.K. Dewett, Modern Economic Theory: Micro & Macro Analysis - Orient Book Distributors, New Delhi.
4. V.L. Mote, Managerial Economics - Tata McGraw Hill, New Delhi
5. Gaurav Dutt & Aswani Mahajan, Dutt & Sundaram's Indian Economy - Sultan Chand & Sons.

IHS222 Principles of Management [1-0-0-1]

Course Objectives

- To introduce Profession of Management and help the students gain understanding of the functions and responsibilities of the manager.
- To provide participants tools and techniques to be used in the performance of the managerial job, and enable them to analyze and understand the environment of the organisation

Course Outcomes

- It will help the students to gain understanding of the functions and responsibilities of managers
- It will provide them tools and techniques to be used in the performance of managerial job

Syllabus

Management - Meaning, Definition and Nature; Evolution of Management – Management Thoughts - Early - Modern - Post modern; Contributions of F.W. Taylor - Henry Fayol – Hawthorne Studies-Behavioural School of Management Approach. Levels of Management, Skills required for a manager, Managerial roles. Management Lessons from Indian Philosophy – Vision, Effectiveness, Efficiency and Teamwork.

Functions of Management: POSDCORB; Characteristics of Management.

Planning: -Meaning – nature – importance -Levels of planning. Objectives – setting objectives – Policies – Planning premises, Types of plans - Process of planning - Decision Making; MBO; Principles in Planning

Organising: - Nature-Purpose-Principles-Organisational Structure and types - Departmentation - Centralization vs. Decentralization - Span of control- Delegation of Authority – Principles in Organising – Line Vs Staff Authority – Networking and Virtual Organizations Staffing: - Meaning, Principles in Staffing, Staffing Functions

Directing: - Leadership – Leadership Traits – Leadership Styles – Principles in Directing –Emerging Trends in Management; Management of Creativity & Innovation – Creative Process – Managing E-Business World – Challenges – Management in Globalized Era – Organizational Social Responsibility Control-System and process of Controlling - Requirements for effective control - The Budget as Control Technique - Information Technology in Controlling – Control Techniques- Control and planning- Types of Control– Reporting - Co-ordination; Principles in Control and Co ordination

Textbooks/References

1. K.Aswathapa, " Essential of Business Administration", Himalaya Publishing House
2. Harold Koontz & Heinz Weihrich, "Essentials of Management", Tata McGraw-Hill,1998
3. JAF Stomer, Freeman R. E and Daniel R Gilbert, "Management", Pearson Education, Sixth Edition, 2004.
4. Stephen P. Robbins and Mary Coulter, "Management", Prentice Hall of India, 8th edition.

5. Tripathy PC and Reddy PN, "Principles of Management", Tata McGraw-Hill, 1999.
6. Y.K. Bhusan, "Fundamentals of Business Organisation & Management"; Sultan Chand & Co., New Delhi

SEMESTER V

CSE311 Artificial Intelligence [3-0-2-4]

Course Objectives

- To introduce the student about the principles of AI & techniques and do exercises in the laboratory to increase the subject understanding.

Course Outcomes

- Acquire a thorough knowledge and fundamental concepts and techniques of artificial Intelligence.
- Learn simulating tools and study AI language for problem solving
- To develop and test mini intelligent systems.

Syllabus

Introduction to Artificial Intelligence: Artificial Intelligence (AI), Major Branches of AI, Applications- Characteristics and Fundamental issues for AI problems, Steps to build Artificial intelligence (AI) systems, Intelligent systems, Characteristics of intelligent systems.

Search Techniques: Why Search, Applications of search, Tree and Graph, Search strategies, Complexity of Search.

Knowledge Representation: Knowledge, Characteristics of knowledge representation, Types of knowledge representation, Propositional Logic, Tautology and Contradiction, Predicate Logic, Production Systems, Semantic network, Frame systems, Scripts.

Neural Networks: Introduction to Neural network, Structure of Neural network.

Structure of Neural network, Neural Network Architecture, Network Layers.

Neural Network Learning, Back-Propagation Algorithm.

Intelligent agents: Introduction to Agents, Functions, Examples of Agents Intelligent Agent classification, Features of intelligent agents, Structure of Agents, Intelligent Agents Models Fuzzy logic: Crisp logic, Fuzzy logic, Membership function, Fuzzy logic Applications.

Expert Systems: What is Expert system, Conventional systems vs. Expert systems, Basic Concepts, Human Expert Behaviors, Knowledge Types, Inferencing, Rules, Structure of Expert Systems, ES Components, Knowledge Engineer, Expert Systems Working, Problem Areas Addressed by Expert Systems, benefits-limitations- Applications of expert systems.

Textbooks/References

1. Stuart J Russell, Peter Norving. Artificial Intelligence: A Modern approach, Third Edition, 2015.
2. Elaine Rich and Kevin Knight, Introduction to Artificial Intelligence, McGraw Hill, Third Edition, 2017.
3. Michael Negnevitsky, Artificial Intelligence: A guide to Intelligent Systems, Addison Wesley, Third Edition, 2017.
4. G.F. Luger, and W.A. Stubblefield, Artificial Intelligence: Structures and Strategies for Complex Problem Solving, Addison-Wesley Publishing Company, 2011.
5. C.S. Krishnamoorthy and S. Rajeev, Artificial Intelligence and Expert Systems for Engineers by CRC Press, 1996.

ECS313 VLSI Design [3-0-2-4]

Course Prerequisites

Semiconductor devices, digital electronics.

Course Objectives

- Study the fundamentals of CMOS circuits and its characteristics.
- Learn the design and realization of combinational & sequential digital circuits.
- Architectural choices and performance tradeoffs involved in designing and realizing the circuits in CMOS technology are discussed

Course Outcomes

The learning outcomes for this course are as follows:

- Be able to create models of moderately sized CMOS circuits that realize specified digital functions.
- Have an understanding of the characteristics of CMOS circuit construction and the comparison between different state-of-the-art CMOS technologies and processes.
- Be able to complete a significant VLSI design project having a set of objective criteria and design constraints.

Syllabus

Introduction to IC Technology – MOS, PMOS, NMOS, CMOS & BiCMOS Technologies.

VLSI Design Flow, MOS Layers, Stick Diagrams, Design Rules and Layout, CMOS Design rules for wires, Contacts and Transistors Layout Diagrams for NMOS and CMOS Inverters and Gates, Scaling of MOS circuits.

CMOS logic: PMOS, NMOS and CMOS, MOS transistors as a switch, CMOS inverter, Power Consumption: Static and Dynamic.

Designing Combinational Logic gates in CMOS: Static CMOS Design, Complementary CMOS, Ratioed Logic, Pass transistor logic, Dynamic CMOS design (Precharge and Evaluation) (Combinational circuits like NAND, NOR and other simple logics using CMOS).

Designing Sequential Logic Circuits: Timing Metrics for sequential Circuits, Setup time, hold time, propagation delay, Latches Vs Registers, Classification of Memory Elements

Datapath subsystem design: Adders, ALU, multipliers Implementation Strategies and testing for Digital ICs: Full Custom, Semicustom, Gate Array based design approaches, Programmable Logic Devices, Field Programmable Gate Arrays, Standard Cell based design, VLSI Design Flow Design for Testability: Ad Hoc Testing, Scan Design, BIST, IDDQ Testing, Boundary Scan.

Textbooks/References

1. Jan M Rabaey, Anantha Chandrakasan, Borivoje Nikolic. Digital Integrated Circuits: A design perspective
2. Michael John Sebastian Smith. Application Specific Integrated Circuits.
3. CMOS logic circuit Design - John .P. Uyemura, Springer, 2007.

4. Weste & Harris, CMOS VLSI Design: A Circuits and Systems Perspective, 4rd ed, Addison Wesley, 2010

IEG314 Digital Signal Processing and its Applications [3-0-2-4]

Course Objectives

- Understand the frequency domain sampling and reconstruction of discrete time signals.
- Study the properties and the development of efficient algorithms for the computation of DFT.
- Study the different windows used in the design of FIR filters and design appropriate filters based on the specifications.
- Learn the procedures to design of IIR filters from the analog filters using bilinear transformation.
- Realization of FIR and IIR filters in different structural forms.
- Understand the architecture and working of DSP processor

Course Outcomes

- Determine response of LTI systems using time domain and DFT techniques.
- Compute DFT of real and complex discrete time signals.
- Computation of DFT using FFT algorithms and linear filtering approach.
- Design and realize FIR and IIR digital filters
- Understand the DSP processor architecture

Syllabus

DISCRETE FOURIER SERIES & FOURIER TRANSFORMS: DFS representation of periodic sequences, Properties of discrete Fourier series, Discrete Fourier transforms: Development of DFT from DTFT, DFT as a linear transformation, Properties of DFT, linear filtering methods based on DFT, Fast Fourier transforms (FFT) - Radix-2 decimation in time and decimation in frequency FFT Algorithms, Inverse FFT.

DESIGN OF FIR DIGITAL FILTERS & REALIZATIONS: Characteristics of linear phase FIR filters and frequency response. Design of linear phase FIR filters using windows – Rectangular, Bartlett, Hanning and Hamming windows. Design of FIR filters using frequency sampling method. Basic structures of FIR systems - Direct form, Cascade form and Lattice structures.

DESIGN OF IIR DIGITAL FILTERS & REALIZATIONS: Analog filter approximations – Butter worth and Chebyshev, Design of IIR Digital filters from analog filters using Lowpass prototype transformation, Bilinear Transformation and Frequency Warping, Bilinear Transformation Design Procedure, Digital Butterworth Filter Design using Bilinear Transformation. Analog and Digital frequency transformations. Basic structures of IIR systems in Direct form I and II.

DIGITAL SIGNAL PROCESSORS: DSP Architecture, DSP Hardware Units, Digital Signal Processors and Manufacturers,

Lab Practice

- Verification of sampling theorem.
- Linear and circular convolution of two given sequences.
- Verification of commutative, distributive and associative properties of convolution.
- Auto and cross correlation of two sequences and verification of their properties.
- Solving a given difference equation.
- Computation of N point DFT of a given sequence and to plot magnitude and phase spectrum
- Verification of DFT properties (like Linearity and Parseval's theorem, etc.)
- DFT computation of square pulse and Sinc function.
- Design and implementation of Low pass and High pass FIR filter to meet the given specifications (using different window techniques) and test the filter with an audio file. Plot the spectrum of audio signal before and after filtering.
- Design and implementation of a digital IIR filter (Low pass and High pass) to meet the given specifications and test with an audio file. Plot the spectrum of audio signal before and after filtering.

Textbooks/References

1. Proakis & Monalakis, "Digital signal processing Principles Algorithms & Applications", 4th Edition, Pearson education, New Delhi, 2007. ISBN: 81-317-1000-9.
2. Li Tan, Jean Jiang, "Digital Signal processing Fundamentals and Applications", Academic Press, 2013, ISBN: 978-0-12-415893.
3. Sanjit K Mitra, "Digital Signal Processing, A Computer Based Approach", 4th Edition, McGraw Hill Education, 2013.
4. Oppenheim & Schaffer, "Discrete Time Signal Processing", PHI, 2003.

IMA312 Information Theory and Coding [3-0-0-3]

Course Objectives

- To equip students with the basic understanding of the fundamental concept of entropy and information as they are used in communications.
- To enhance knowledge of probabilities, entropy, measures of information.
- To guide the student through the implications and consequences of fundamental theories and laws of information theory and coding theory with reference to the application in modern communication and computer systems

Course Outcomes

- Calculate the information content of a random variable from its probability distribution.
- Relate the joint, conditional, and marginal entropies of variables in terms of their coupled probabilities.
- Define channel capacities and properties using Shannon's Theorems.
- Construct efficient codes for data on imperfect communication channels.
- Generalize the discrete concepts to continuous signals on continuous channels.

Syllabus:

Entropy, Relative Entropy Mutual information and its properties, entropy for discrete ensembles, Shannon's source coding theorem, Encoding of discrete sources, Kraft Inequality, Huffman Codes, Shannon Fano Elias Coding, Arithmetic coding, Lempel Ziv Welch Coding

Channel capacity, Capacity of binary symmetric channel, erasure channels and other discrete channels, Shannon's noisy coding theorem and converse for discrete channels, Differential entropy, Application to continuous channels: Gaussian channel, Bandlimited channel, channels with coloured noise.

Introduction to Galois field, addition and multiplication tables of Galois field elements, Linear block codes, Error detection and error correction capabilities of Linear block codes, parity check codes, Hamming codes, encoding and decoding of linear codes, syndrome decoding of linear codes, Cyclic codes, Convolutional codes, Graphical approach: state, tree and trellis diagram, Viterbi decoding; Maximum likelihood decoding; Introduction to iterative codes and its sub-optimal decoding algorithms, BCH codes, RS codes, LDPC Codes.

Textbooks/References

1. Elements of Information Theory, T.M. Cover and J.A.Thomas, Wiley 19914
2. R. W.Yeung, A First Course in Information Theory, Kluwer Academic Publisher, 2002.
3. R. E. Blahut, Algebraic Codes for Data Transmission, Cambridge University Press, 2003.
4. S. Lin and D. J. Costello, "Error Control Coding," 2nd Ed., Pearson Prentice Hall, 2004, ISBN-13: 978130426727.

ECE312 Microwave Engineering [3-1-0-4]

Course Prerequisites:

Electromagnetic theory.

Course Objectives

- To inculcate understanding of the basics required for microwave circuits.
- To deal with the issues in the design of microwave amplifier.

- To instill knowledge on the properties of various microwave components.
- To deal with the microwave generation and microwave measurement techniques.

Course Outcomes

Upon completion of the course, students will be able to:

- Explain the active & passive microwave devices & components used in Microwave communication systems.
- Generate Microwave signals and design microwave amplifiers.

Syllabus

Review of electromagnetics: Maxwell's equations, plane wave solutions; Transmission lines: coaxial lines, rectangular waveguides, Microstrip; Network analysis: scattering matrix, transmission matrix formulations; Matching networks: Lumped element designs and limitations, single and double-stub tuned designs, Quarter wavelength transformers, multi-section matching transformers; Smith chart, Analysis of modes TE, TM, TEM.

Active microwave circuit design: characteristics of microwave transistors, mixers and detectors, Oscillators; Amplifier design: LNA and Power amplifiers, gain and stability, design for noise figure, Single-stage amplifier design; Radar and microwave generators.

Introduction to antenna, Types of antennas, Noise in microwave circuits: dynamic range and noise sources, equivalent noise temperature, system noise figure considerations

Textbooks/References

1. David M. Pozar, 'Microwave Engineering,' 3rd. ed., John Wiley & Sons, 2005.
2. Guillermo Gonzalez, Microwave Transistor Amplifiers, 2nd. ed., PrenticeHall, 1997.
3. Thomas H. Lee, Planar Microwave Engineering: A Practical Guide to Theory, Measurement, and Circuits, 1 st Edition, Cambridge University Press, 2004
4. Robert E Colin, "Foundations for Microwave Engineering", John Wiley & Sons Inc, 2005

IHS311 Human Resource Management [1-0-0-1]

Course Objectives

- To provide participants a synthesized framework of Human Resources theory & Practice.
- To impart fundamentals of HR Practices in Organizations.
- Learn to align HR systems with the strategic business objectives of a firm.

Course Outcomes

- Students will become familiar with the fundamentals of HR

- It will equip them the tools and techniques of modern Human Resources Management.

Syllabus

Introduction to Human Resource Management-Importance-Scope and Objectives. Evolution. Line and Staff aspects of HRM, Line managers; Duties of Human Resources Managers-Human capital management.

Job analysis: Methods for collecting Job Analysis Information, Human Resource Planning and Recruiting: The Recruitment and Selection process- Planning and Forecasting, Internal and External sources of candidates, Writing Job Description & Job Specification, Managing HR in challenging times- Testing and Selection: Basic testing concepts, Types of Tests. Interview: Process and Types, Guidelines for Interviews.

Training & Performance Management: Orientation, Training Process, Training Needs Analysis, Training Techniques- On -the-Job & Off-the -Job Training Methods, OJT Process, Training Evaluation. Management Development Programs: Case Study and other Modern Training Methods. Performance Management & Appraisal: Process and Techniques. Career Planning and Management Concepts.

Compensation

Establishing Pay Rates: Steps, Job Evaluation -Wage and Salary administration- Steps and factors affecting, Incentives Benefits and services: Statutory Benefits - Non-statutory Benefits - Insurance Benefits -Retirement Benefits, Flexible Benefits Programs. ESOPs, QWL.

Industrial Relations & Trends in HR

Industrial relations: Significance, Objectives, Approaches. Industrial Disputes- Causes, Forms, Preventive Machinery. Collective Bargaining: Basic Concepts. Long term settlements: Cases in India. Trade unions: Definition, Objectives, Functions Social Security in India, Employee welfare, Grievance Handling and Discipline-Sources and forms of Grievances -Grievance Procedure, Disciplinary Procedure. Participative Decision-making process - Role of quality circle in TQM. Strategic Human Resources Management, Strategic HRM tools. An over view of HR Analytics.

Textbooks/References

IHS312 Financial Management and Accounting [1-0-0-1]

Course Objectives

- To familiarize participants with fundamentals of Financial Management in an Organization.
- To provide the participants various techniques in Financial Management.
- To give an overview of the emerging financial issues facing an Organization.

Course Outcomes

- Students will become familiar with accounting standards and financial management.
- It will equip the skills for better financial management.

Syllabus

Introduction to Finance: Introduction to Financial management: Business Finance- Concept, types and scope. Financial management: objectives, functions and scope - Interface of financial management with other functional areas. Role of finance manager- Financial forecasting - Financial planning. Risk and Return concept: - Relationship between risk and return - Risk Diversification.

Time Value of Money & Investment Decisions: Process of compounding - Process of discounting - Future value of Single cash flow and annuity - Present value of a single cash flow and annuity. Investment Decisions: - Capital budgeting - Process of capital budgeting - selection of projects - Estimation of cash flows - Payback and Discounted payback period - Accounting rate of return- NPV - IRR - Capital Budgeting decisions under risk - Capital Rationing - Project selection under rationing.

Finance Decisions: Financing and Capital Structure Decision: Sources of Finance: External and Internal financing. Cost of different sources of capital -Weighted average cost of capital (WACC) and Marginal cost of capital. Capital structure decisions - meaning and pattern- Theories of capital structure- Net income approach - Net operating income approach- Traditional approach-MM approach - Optimum capital structure. Leverage - operating, financial and composite leverage.

Dividend Decisions: Dividend policy - dividend and its forms - objectives of dividend policy- relevance and irrelevance. Theories of dividend decisions: Walter's Approach - Gordon's Approach - MM Approach

Working Capital: Management of Working capital: Meaning and Need of Working capital - factors affecting composition of working capital - Inter dependence among components of working capital - Estimation of working capital - Cash management- Cash flow statement and fund flow statement- Receivables management.

Textbooks/References

1. Brealey, Richard A and Stewart CMyers. Principles of Corporate Finance. McGraw Hill India, 2012.
2. Chandra Prasanna, Financial Management- Theory & Practice, Tata McGraw Hill, 2014.
3. James C Vanhorne, John M WachowiczJr, Fundamentals of Financial Management, Pearson Education Limited, New Delhi.
4. Lawrence J Gitman, Principles of Managerial Finance, Pearson Education limited. New Delhi.
5. Pandey IM, Financial Management. Vikas Publishing House, 2009.
6. Reddy, G Sudarsana, Financial Management, Himalaya Publishing House, 2011.
7. Van Horne James, Financial Management Policy, Prentice Hall India

IHS313 Operations and Supply Chain Management [1-0-0-1]

Course Objectives

- To gain a working understanding of logistics principles and to expose students to the basic concepts the language of logistics and supply chain management.
- To refine applied data analysis skills by analyzing and using supply chain data to evaluate supply chain performance and to make business decisions.
- To understand current challenges faced by supply chain professionals and to provide a basis for thinking through these challenges.

Course Outcomes

- Students will become familiar with basic concepts of Supplychain Management.
- Students will get a better understanding on web based and electronic applications of supplychain management.

Syllabus

Introduction to Production & Operations Function:- Scope of Production and Operations Management - Interaction of Operations Management with other functional areas of Management - Manufacturing and Non Manufacturing operations and their Classifications - Operations Strategy: Elements of Operations Strategy - 5Ps of Operations.

Production & Operations Planning, Manufacturing System, Layout: Production & Operations Planning and Control, Role of Production Planning & Control in Operations Management - Plant Location & Layout: Steps in location selection - Factors influencing Layout - Principles of Layout - Layouts by Products and Process - Hybrid Layout -Design of Operations Systems: Aggregate planning and Master Scheduling, MRP, CRP. Line Balancing & Sequencing - Capacity Planning

Materials Management & Vendor Management :Material Handling: Material Handling Principles - Types - Selection & Design of Material Handling System. Materials Management - Functions - Material planning and Budgeting - Value Analysis - Purchase functions and Procedure - Inventory control - Types of Inventory - Safety stock - Inventory Control Systems - Economic Order Quantity (EOQ) - Perpetual - Periodic - Just In Time (JIT) - Managing Vendors; Vendor Analysis, Rating and Selection - Procedure and Criteria.

Work Study & Maintenance: Work study, Time and Method study: Definition - Importance - Aims and Procedures - Implications on Productivity - Work measurement - Work sampling - Work environment - Industrial safety - Value analysis. Basics of Maintenance Management - Maintenance Decisions

Supply Chain Management & Lean Systems: Supply Chain Management -Concept of Supply chain, Stages and flows in Supply chain, Terminology in Supply chain management -Supply chain disruption- Bull Whip effect. Lean Systems - Basic understanding about Lean concepts- Pull and Push systems, Jidoka, Poke-Yoke, 5S, Total Preventive Maintenance (TPM), Toyota Production System, Kanban System.

Textbooks/References

1. Lee J. Krajewski et al, Operations Management, Process and Supply chains. 11th Edition Pearson India Education Services Ltd. India.

2. Russel & Taylor, Wiley, Management, Quality and Competitiveness in a Global Environment, Fifth Edition, India Edition
3. BuffaSarin, Wiley, Modern Production and Operations Management, India Edition
4. KanishkaBedi, Production and Operations Management, Oxford University Press.
5. Aswathappa K and ShridharaBhat K, Production and Operations Management, Himalaya Publishing House, Revised Second Edition, 2008.
6. Pannerselvam R, Production and Operations Management, Prentice Hall India, Second Edition, 2008.
7. Mahadevan B, Operations Management Theory and Practice, Pearson Education, 2007.

SEMESTER VI

ECE323 Internet of Things [3-0-2-4]

Course Prerequisites

The participants should have prior knowledge on the following topics/courses:

- Computer Networks
- Digital Design and Electric Circuits.
- Fundamentals of Programming

Course Objectives

- To introduce the concept and the basics of IoT technologies,
- To provide knowledge on various applications of IoT based technologies and their associated circuits,
- To enable awareness on the different products that were designed based on IoT.

Course Outcomes

- The students will have knowledge on various applications of IoT.
- The students will have practical knowledge on operating with sensors/actuators.
- The students will be capable to design their own IoT based applications using Arduino or Raspberry PI boards.

Syllabus

Introduction - Sensor basics, sensing and actuation, basics of networking - wired, wireless, MANET, PAN, wireless and wired protocols. Communication protocols- IEEE standards, 5G era, sensor communications, connectivity challenges, fading and attenuation.

IoT architectures and programming – basic architectures, Data processing mechanisms, scalability issues, visualization issues, analytics basics, utility of cloud computing, fog computing, and edge computing, advanced IoT architectures Raspberry Pi and Arduino programming – Golang and Nodejs programming.

Applications - IoT for societal upliftment, industrial automation (Industry 4.0), smart city, smart home, smart transportation, smart healthcare, smart agricultures, golang based implementation. **Text Book/References**

1. Pethuru Raj and Anupama C. Raman, The Internet of Things: Enabling Technologies Platforms, and Use Cases, CRC Press, First edition, 2017.
2. Honbu Zhou, The Internet of Things in the Cloud: A Middleware Perspective, CRC press, First edition, 2012.
3. Arshdeep Bahga and Vijay Madisetti, Internet of Things: A Hands-on Approach, Universities Press, First edition, 2014.
4. Mung Chiang, Bharath Balasubramanian, Flavio Bonomi, Fog for 5G and IoT (Information and Communication Technology Series, Wiley series, First edition, 2017.
5. Alan A. A. Donovan, Brian W. Kernighan, The Go Programming Language, Addison- Wesley Professional Computing Series, First edition, 2015.

ECE322 Embedded Systems [3-0-2-4]

Course Prerequisites

Digital Electronics, Microprocessors.

Course Objectives

This course will enable students to:

- Explain the architectural features and instructions of 32 bit microcontroller ARM Cortex M3.
- Develop Programs using various instructions of ARM Cortex M3 and C language for different applications.
- Understand the basic hardware components and their selection method based on the characteristics and attributes of an embedded system.
- Develop the hardware software co-design and firmware design approaches.
- Explain the need of real time operating system for embedded system applications.

Course Outcomes

After studying this course, students will be able to:

- Describe the architectural features and instructions of 32 bit microcontroller ARM Cortex M3.
- Apply the knowledge gained for Programming ARM Cortex M3 for different applications.
- Understand the basic hardware components and their selection method based on the characteristics and attributes of an embedded system.
- Develop the hardware software co-design and firmware design approaches.
- Explain the need of real time operating system for embedded system applications.

Syllabus

Introduction to embedded systems and embedded processors. Embedded system architecture, classifications of embedded systems, challenges and design issues, CISC vs. RISC.

ARM-32 bit Microcontroller: Thumb-2 technology and applications of ARM, Architecture of ARM Cortex M3, Various Units in the architecture, Debugging support, General Purpose Registers, Special Registers, exceptions, interrupts, stack operation, reset sequence.

ARM Cortex M3 Instruction Sets and Programming: Assembly basics, Instruction list and description, Thumb and ARM instructions, Special instructions, Useful instructions, CMSIS.

Embedded System Components: Embedded Vs General computing system, Classification of Embedded systems, Major applications and purpose of ES. Elements of an Embedded System (Block diagram and explanation), Differences between RISC and CISC, Harvard and Princeton, Big and Little Endian formats, Memory (ROM and RAM types), Sensors, Actuators, Optocoupler, Communication Interfaces (I2C, SPI, IrDA, Bluetooth, Wi-Fi, Zigbee only).

Embedded System Design Concepts: Characteristics and Quality Attributes of Embedded Systems, Operational and non-operational quality attributes, Embedded Systems-Application and Domain specific, Hardware Software Co-Design and Program Modeling, Embedded firmware design and development. RTOS and IDE for Embedded System Design: Operating System basics, Types of operating systems, Task, process and threads (Only POSIX Threads with an example program), Thread preemption, Preemptive Task, scheduling techniques, Task Communication, Task synchronization issues – Racing and Deadlock, Concept of Binary and counting semaphores (Mutex example without any program), How to choose an RTOS, Integration and testing of Embedded hardware and firmware, Embedded system Development Environment – Block diagram (excluding Keil), Disassembler/decompiler, simulator, emulator and debugging techniques.

Textbooks/References

1. Embedded systems Architecture, Programming and design, Raj Kamal, second Edition, Tata MC Graw-Hill, 2008
2. Designing embedded systems with PIC microcontrollers: Principles and Applications, Tim Wilmshurst, Second Edition, Elsevier, 2005.
3. Embedded System design, Steve Heath, Second Edition, Newnes, 2002.
4. W.A. Smith, "ARM Microcontroller Interfacing: Hardware and Software, Eketor, 2010.
5. Joseph Yiu, "The Definitive Guide to the ARM Cortex-M3", 2nd Edition, Newnes, (Elsevier), 2010.
6. Steve Furber, "ARM System Architecture", Edison Wesley Longman, 1996.
7. William Hohl, "ARM Assembly Language Fundamentals and Techniques ", CRC Press, 2009.

IOEXXX Advanced Computer Architectures [3-0-0-3]

Course Objectives

The objective of the course is

- To provide in-depth coverage of current and emerging trends in computer architecture focusing on performance and the hardware/software interface.
- On analyzing fundamental issues in architecture design and their impact on application performance.
- Exploring their own interests in custom projects and assignments.

Course Outcomes A student who has successfully completed this course should be able to:

- Analyze various performance characteristics of a computer system.

- Apply digital design techniques to the microarchitecture construction of a processor.
- Translate assembly language programs to/from high-level language codes and algorithms.
- Analyze hardware & software trade-offs to design the instruction set architecture (ISA) interface.
- Understand advanced issues in design of computer processors, caches, and memory.
- Analyze performance trade-offs in computer design.

Syllabus

Introduction to Computer Design and Quantitative Principles of Architecture Performance Analysis : Technology and computer trends, Measuring computer system performance, Benchmarks and metrics, CPI, Moore's law, Amdahl's Law.

Instruction Set Principles and Examples: Classification of Instruction Set Architectures (ISA) - RISC, CISC, VLIW, EPIC, Predicated execution and compiler-controlled speculation

Advanced Microarchitecture and Instruction Level Parallelism : Superscalar and pipeline operation, Instruction-Level Parallelism (ILP)

Memory-Hierarchy Design: Multi-level cache design issues , Performance evaluation.

Architecture Implementation Issues and Analysis: Power- Dynamic Voltage Frequency Scaling (DVFS), Energy-Delay Product (EDP), Architecture physical layer concepts including device & layout, manufacturing constraints, architectures, defect tolerance

Textbooks/References:

1. Hennessy and Patterson, Computer Architecture- A Quantitative Approach, 4th or later Edition (ISBN-13: 978-0123704900 ISBN-10: 0123704901 Edition: 4th).
2. Kai Hwang and Naresh Jotwani, Advanced Computer Architecture (SIE): Parallelism, Scalability, Programmability, McGraw Hill Education 3/e. 2015.

IOEXXX Computer Vision [3-0-0-3]

Course Objectives

- To introduce students the fundamentals of image formation;
- To introduce students the major ideas, methods, and techniques of computer vision and pattern recognition;
- To develop an appreciation for various issues in the design of computer vision and object recognition systems
- To provide the student with programming experience from implementing computer vision and object recognition applications.

Course Outcomes

After completing the course you will be able to:

- Identify basic concepts, terminology, theories, models and methods in the field of computer vision,

- Describe known principles of human visual system,
- Describe basic methods of computer vision related to multi-scale representation, edge detection, detection of other primitives, stereo, motion and object recognition.

Syllabus

Introduction to computer vision, geometric camera models, light and shading, local image

features: SIFT, HOG, texture and shape descriptors, active contour, segmentation, deformable models, RANSAC, image registration, learning and classification strategies, image classification, object detection and recognition, stereopsis, tracking, applications.

Textbooks/References

1. Forsyth and Ponce, "Computer vision: a modern approach," 2nd Ed., Pearson, 2012.
2. Sonka, Hlavac and Boyle, "Digital image processing and computer vision," Cengage learning, 2008.
3. Rick Szeliski, "Computer Vision: Algorithms and Applications," Springer, 2011.

IOEXXX Wireless Sensor Networks [3-0-0-3]

Course Objectives

- To Understand the basic WSN technology and supporting protocols, with emphasis placed on standardization basic sensor systems and provide a survey of sensor technology
- Understand the physical layer and MAC layer design issues in WSN and the techniques to address them
- Understand the medium access control protocols and address physical layer issues
- Learn key routing protocols for sensor networks and main design issues
- Learn the significance of localization techniques and its implementation
- Learn transport layer protocols for sensor networks, and design requirements

Course Outcomes

- Have knowledge and understanding of basic WSN technology and supporting protocols and Technology
- Have knowledge and to Identify medium access control protocols and address physical layer issues
- Have knowledge routing protocols for sensor networks and main design issues
- Have knowledge of transport layer protocols for sensor networks, and design requirements
- Identify the design issues in physical and MAC layers of WSN and the techniques to address them
- Design and develop routing protocols and localization techniques

- Understand the application environment and OS requirements in WSN

Syllabus

Introduction, WSN Architecture and Protocol Stack: Physical Layer, Data Link Layer, Network Layer, Transport Layer, Application Layer, WSN Applications, Factors Influencing WSN Design: Hardware Constraints, Fault Tolerance, Scalability, Production Costs, WSN Topology, Transmission Media, Power Consumption. Physical Layer: Physical Layer Technologies, RF, Other Techniques, Overview of RF Wireless Communication, Wireless Channel Effects, Attenuation, Multi-path Effects, Channel Error Rate, Unit Disc Graph vs. Statistical Channel Models, PHY Layer Standards, IEEE 802.15.4 Medium Access Control: Challenges for MAC, CSMA Mechanism, Contention-Based Medium Access-S-MAC, B-MAC, CC-MAC, Other Contention-Based MAC Protocols, Reservation-Based Medium Access, Hybrid Medium Access Network Layer: Challenges for Routing, Data-centric and Flat-Architecture Protocols, Hierarchical Protocols, Geographical Routing Protocols, QoS-Based Protocols Localization: Challenges in Localization, Ranging Techniques, Range-Based Localization Protocols, Range-Free Localization Protocols, Topology Management: Deployment, Power Control, Activity Scheduling, Clustering Hardware Design for WSNs: General Wireless Sensor Node Architecture, System-on-Chip and Component-based Design, Design Guidelines -Microcontroller Selection, Communication Device Selection, Sensing Device Design, Power Supply Device Design, Design Case - Temperature Sensor Design, CO Sensor Design, Sensor Node Circuit Design, Power Management, Energy Scavenging, Solar Energy Harvesting Unit, Maximum Power Point Tracking Unit, Power Management Unit, Design Case.

Application support: Data Aggregation, In-network processing, Security issues, Operating Systems and Execution Environment, Future Trends in Wireless Sensor Networks.

Textbooks/References

1. Wireless sensor networks, Akyildiz, Ian F., and Mehmet Can Vuran, John Wiley & Sons, 2010.
2. Wireless sensor networks, Yang, Kun, Springer 2014.
3. Protocols And Architectures for Wireless Sensor Networks, Holger Karl and Andrea Willig, John Wiley & Sons, 2018.
4. Adhoc Wireless Networks: Architectures and Protocols, C Siva Ram Murthy and B S Manoj, Prentice Hall India.
5. Wireless Sensor Networks: A Networking Perspective, Jun Sheng and Abbas Jamalipour, John Wiley & Sons.

IOEXXX Cloud Computing [3-0-0-3]

Course Objectives

The main objectives of this course are:

- To gain knowledge on virtualization techniques.
- To frame VM clusters.
- To migrate or consolidate VM machines.

- To understand the working methodology of existing clouds, such as, Amazon, Opennebula, and so forth.
- To learn how to program clouds using new programming models.

Course Outcomes

- The students will learn the basic technologies for cloud.
- Apply appropriate cloud services for their applications.
- Design cloud services using goLang or nodejs.
- Learn how to program public clouds such as AWS or GCE.

Syllabus

Base Technologies - Review: Introduction, Grid Computing, Cluster, P2P computing, and so forth, System Models for Distributed and Cloud computing.

Virtualization: Virtualization concepts, levels of Virtualization, VM Tools and mechanisms,

Virtualization of CPU, Memory, and I/o devices, VM server consolidation, VirtualBox,

VMWare Vsphere - Datacenter Automation.

Cloud Infrastructure / Architectures: Design Challenges of Clouds, Public cloud platforms.

GCE, AWS, Azure, Resource Management in Clouds, cloud environments - openstack and opennebula, security aspects of clouds, Storage aspects of clouds, introduction to programming models.

Advanced topics: Kubernetes, Docker containers, DevOps, Cloud Networking - SDN, HPC in cloud, IoT cloud, Microservices.

Lab Practice

VMs using virtualbox, VMs using AWS, server automation using VMware Cloud, cloud services, Kubernetes, OpenStack or OpenNebula.

Textbooks/References

1. Kai Hwang, Geoffrey C. Fox, Jack K. Dongarra, Distributed and Cloud Computing: From parallel processing to Internet of Things, Morgan Kaufmann 2013.
2. William Stallings, Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud, Pearson publishers, 2016.
3. Jonathan Baier, Getting Started with Kubernetes: 2nd Edition, Packt publishers, 2015.
4. Hideto Saito, Hui-Chuan Chloe Lee, and Cheng-Yang Wu, DevOps with Kubernetes Accelerating software delivery with container orchestrators, Packt publishers, 2017.
5. Gigi Sayfan, Mastering Kubernetes: Large scale container deployment and management, Packt publishers, 2016.
6. Kevin Hoffman and Dan Nemeth, Cloud Native Go: Building Web Applications and Microservices for the Cloud with Go and React (Developer's Library), Pearson publishers, 2016.

7. Bob Familiar, Microservices, IoT, and Azure: Leveraging DevOps and Microservice Architecture to deliver SaaS Solutions, Apress publishers, 2015.
8. Dirk Slama, Frank Puhlmann, Jim Morrish, and Rishi M. Bhatnagar, Enterprise IoT: Strategies and Best practices for Connected products and services, O’Rielly publishers, 2015.

IOEXXX System-On-Chip Design [3-0-0-3]

Course Objectives

- Provide a comprehensive introduction to SoC technology.
- Give an overview to SoC design challenges and design flow.
- Analyze hardware/software tradeoffs, algorithms, and architectures to optimize the system based on requirements and implementation constraints.
- Analyze tradeoffs and explore architecture and micro-architecture design spaces to develop and synthesize custom hardware accelerators

Course Outcomes

On completion of this course the students will be able to:

- Design processors keeping area, power, and speed as constraints
- Analyze memories using reconfigurable architectures for rapid prototyping
- Analyze system on chip and board based systems.
- Design custom hardware accelerators

Syllabus

System Approach and Chip Basics: System Architecture, Components of the System, Hardware and Software. An approach for SoC Design, System Architecture and Complexity. Chip Basics. Cycle Time, Die Area and Cost, Ideal and Practical Scaling, Power, Area–Time–Power Trade-Offs in Processor Design, Reliability, Configurability. Memory Design: System-on-Chip and Board-Based Systems – Scratchpads and Cache Memory, Basic Notions, Cache Organization, Cache Data, Write Policies, Strategies for Line Replacement at Miss Time, Other Types of Cache, Split I- and D-Caches and the Effect of Code Density, Multilevel Caches, Virtual-to-Real Translation, SoC (On-Die) Memory Systems, Board-based (Off-Die) Memory Systems, Simple DRAM and the Memory Array, Models of Simple Processor–Memory Interaction. Full custom and semicustom ASIC designs, FPGAs. Architecture and micro-architecture design spaces to develop and synthesize custom hardware accelerators

Textbooks/References

1. Micheal J Flynn and Wayne Luk, “Computer System Design: System-on-Chip,” Wiley, First Edition, 2011.
2. M.J.S .Smith, “Application Specific Integrated Circuits” – Pearson Education, 2003.
3. Wayne Wolf, “Modern VLSI Design – System – on – Chip Design”, Prentice Hall, 3rd Edition 2008.

ECSXXX Advanced Communication Networks [3-1-0-4]

Course Objectives

This is a graduate level course on advanced computer communication and networking technologies. Major objectives are:

- To understand the protocol layering and physical level communication.
- To analyze the performance of a network.
- To understand the various components required to build different networks.
- To learn the functions of network layer and the various routing protocols.
- To familiarize the functions and protocols of the Internet servers

Course Outcomes

Upon successful completion of this course, the student will be able to:

- Understand the basic layers and its functions in Communication networks.
- Design and develop protocols for Communication Networks.
- Understand the basic mechanisms in Quality of Service in networking.
- Analyze and design routing algorithms.
- Design protocols for various functions in the network

Syllabus

Overview of Internet-Concepts, challenges and history. Overview of - ATM. TCP/IP Congestion and Flow Control in Internet-Throughput analysis of TCP congestion control. TCP for high bandwidth delay networks. Fairness issues in TCP. Real Time Communications over Internet, Adaptive applications. Latency and throughput issues, Integrated Services Model (intServ), Resource reservation in Internet, RSVP, Characterization of Traffic by Linearly Bounded Arrival Processes (LBAP), Leaky bucket algorithm and its properties. Packet Scheduling Algorithms requirements and choices, Scheduling guaranteed service connections. GPS, WFQ and Rate proportional algorithms. High speed scheduler design. Theory of Latency Rate servers and delay bounds in packet switched networks for LBAP traffic.; Active Queue Management - RED, WRED and Virtual clock. Control theoretic analysis of active queue management. IP address lookup-challenges. Packet classification algorithms and Flow Identification Grid of Tries, Cross producting and controlled prefix expansion algorithms. Admission control in Internet. Concept of Effective bandwidth. Measurement based admission control. Differentiated Services in Internet (DiffServ). DiffServ architecture and framework. IPV4, IPV6, IP tunnelling, IPswitching and MPLS, Overview of IP over ATM and its evolution to IP switching. MPLS architecture and framework. MPLS Protocols. Traffic engineering issues in MPLS.

Textbooks/References

1. Jean Wairand and Pravin Varaiya, "High Performance Communications Networks", 2 nd edition, 2000.
2. Jean Le Boudec and Patrick Thiran, "Network Calculus A Theory of Deterministic Queueing Systems for the Internet", Springer Veriag, 2001.
3. Zhang Wang, "Internet QoS", Morgan Kaufman, 2001.
4. Anurag Kumar, D. Manjunath and Joy Kuri, "Communication Networking: An Analytical Approach", Morgan Kaufman Publishers, 2004.
5. George Kesidis, "ATM Network Performance", Kluwer Academic, Research Papers, 2005

ECSXXX Automotive Electronics [3-1-0-4]

Course Objectives

This course will enable students to:

- Understand the basics of automobile dynamics and design electronics to complement those features.
- Design and implement the electronics that attribute the reliability, safety, and smartness to the automobiles, providing add-on comforts.

Course Outcomes

- Explain the electronics systems used for control of automobiles
- Select sensors, actuators and control systems used in automobiles
- Diagnose the faults in the sub systems and systems used automobile

Syllabus

Automotive Fundamentals Overview -Evolution of Automotive Electronics, Automobile Physical Configuration, Survey of Major Automotive Systems, The Engine - Engine Block, Cylinder Head, Four Stroke Cycle, Engine Control, Ignition System - Spark plug, High voltage circuit and distribution, Spark pulse generation, Ignition Timing, Diesel Engine, Drive Train Transmission, Drive Shaft, Differential, Suspension, Brakes, Steering System, Starter Battery - Operating principle.

The Basics of Electronic Engine Control Motivation for Electronic Engine Control Exhaust Emissions, Fuel Economy, Concept of an Electronic Engine control system, Definition of General terms, Definition of Engine performance terms, Engine mapping, Effect of Air/Fuel ratio, spark timing and EGR on performance, Control Strategy, Electronic Fuel control system.

Analysis of intake manifold pressure, Electronic Ignition. Control Systems - Automotive Control System applications of Sensors and Actuators Typical Electronic Engine Control System, Digital Engine Control Systems

Control Units -Operating conditions, Design, Data processing, Programming, Digital modules in the Control unit, Control unit software.

Automotive Networking - Bus Systems Classification, Applications in the vehicle, Coupling of networks Vehicle Motion Control Typical Cruise Control System, Digital Cruise Control

System, Digital Speed Sensor, Throttle Actuator, Digital Cruise Control configuration, Cruise Control Electronics (Digital only), Antilock Brake System (ABS), Future Automotive Electronic Systems - Alternative Fuel Engines, Electric and Hybrid vehicles, Fuel cell powered cars, Collision Avoidance Radar warning Systems, Low tire pressure warning system, Heads Up display, Speech Synthesis, Navigation - Navigation Sensors - Radio Navigation, Signpost navigation, dead reckoning navigation, Voice Recognition Cell Phone dialing, Advanced Cruise Control, Stability Augmentation, Automatic driving.

Textbooks/References

1. William B. Ribbens, "Understanding Automotive Electronics", 6th Edition, Elsevier Publishing.
2. Robert Bosch GmbH (Ed.) Bosch Automotive Electrics and Automotive Electronics Systems and Components, Networking and Hybrid Drive, 5th edition, John Wiley & Sons Inc., 2007.

ECSXX Micro-Electro-Mechanical Systems [3-1-0-4]

Course Objectives

- To introduce the basic concepts of micro systems and applications of MEMS.
- To introduce the fundamental sensing and actuating principles used in MEMS based structures.
- To introduce the various materials and their properties used for micromachining techniques.
- To analyze the fundamentals of micromachining and micro fabrication techniques.
- To impart knowledge of the basic concept of packaging of MEMS based microdevices.

Course Outcomes

- Understand the basic overview of MEMS and Microsystems with broad category of MEMS & Micro system applications.
- Understand the basic sensing and actuating principles used in MEMS based structures.
- Give an outline of various materials used for MEMS.
- Understand the microsystem fabrication processes and analyze the different micro manufacturing processes.
- Describe the packaging techniques for MEMS devices

Syllabus

MEMS and Microsystems: Introduction components of a microsystem, Multidisciplinary nature of Microsystem design and manufacture. Modern applications of MEMS.

Basic sensing and actuating principles used in MEMS based structures: Micro Sensors Acoustic wave sensors, Biomedical sensors and Biosensors, Pressure sensors, Thermal sensors. Actuation techniques for microdevices - actuation using: thermal

forces, shape memory alloys, piezoelectric crystals, electrostatic forces. MEMS with micro actuators - Micro grippers - micro motors, micro valves, micro pumps, Micro accelerometer, Microfluidics.

Materials for MEMS- substrate and wafers, Silicon - properties, silicon compounds- Silicon Dioxide, Silicon carbide, Silicon Nitride and Polycrystalline Silicon, Silicon Piezoresistors, Gallium Arsenide, Quartz, Piezoelectric Crystals, Polymers- SU-8, PMMA, PDMS, LangmuirBlodgett Films, Packaging materials.

Micro System fabrication - Photolithography, Ion implantation, Diffusion, Oxidation, DepositionChemical Vapor Deposition, Physical Vapor Deposition, Deposition by Epitaxy, Etching. Overview of Micro manufacturing - Bulk micro manufacturing, Surface micro machining, LIGA process.

Micro system Packaging: General considerations in packaging design, Levels of Micro system packaging - die level, device level, system level, Interfaces in microsystem packaging, Essential packaging technologies - die preparation, surface bonding, wire bonding, sealing.

Textbooks/References

1. Tai-Ran Hsu - 'Mems & Microsystems Design and Manufacturing' - John Wiley & Sons - 2008 - 2nd Edition.
2. Marc J Madou - 'Fundamentals of Microfabrication' - CRC Press - 2002 - 2nd Edition.
3. Chang Liu, "Foundations of MEMS", Pearson International Edition, 2006.
4. Introductory MEMS, Thomas M Adams, Richard A Layton, Springer International Publishers.

ECSXXX Biomedical Signal Processing [3-1-0-4]

Course Objectives

- To understand the basic signals in the field of biomedical.
- To study origins and characteristics of some of the most commonly used biomedical signals, including ECG, EEG, evoked potentials, and EMG.
- To understand Sources and characteristics of noise and artifacts in bio signals.
- To understand use of bio signals in diagnosis, patient monitoring and physiological investigation
- To explore research domain in biomedical signal processing

Course Outcomes

After successfully completing the course students will be able to:

- Model a biomedical system.
- Understand various methods of acquiring bio signals.
- Understand various sources of bio signal distortions and its remedial techniques.
- Analyze ECG and EEG signal with characteristic feature points.

- Basic understanding of diagnosing bio-signals and classifying them.

Syllabus

Biomedical Signals: Bioelectric Signals and Electrodes: Bio-potentials and their origin: ECG, EEG, EMG, ENG, ERG, EOG, MEG. Biomedical Instrumentation System, biomedical transducers, electrodes and their characteristics. Origin of bio potentials. Sources and contamination of Noise in bio signals. Classification of biomedical signals.

Cardio Vascular and Nervous System: Cardiovascular system, Coronary and Peripheral Circulation, Electrical Activity of the heart, Lead configurations, ECG data acquisition, ECG recorder, Concept of Blood Pressure Measurement, Nervous System: Structure and functions of Neurons, Electrical activity of nerve cell, Synapse, Reflex action and Receptors. Analysis of Electrical Activity of Heart : ECG signal parameters & their estimation - Use of multiscale analysis for ECG parameters estimation, Noise & Artifacts, ECG Signal

Processing: QRS detection, Highlight the Feature points of ECG and its classification for Normal and Abnormal state using Multilayer Perceptron. Analysis of Electrical Activity of Brain 6L Electroencephalogram - Structure of brain, EEG signal acquisition, EEG rhythms & waveform categorization of EEG activity - recording techniques - EEG applications- Epilepsy, sleep disorders, brain computer interface. Use of Fourier Transform in EEG Signal Analysis.

Analog Signal Processing: Basics of Instrumentation Amplifier, Isolation amplifier, Grounding and shielding techniques. Integer Filters: Basic design Concept, Low Pass and High Pass Filters, Band Pass, Band Stop and Band Reject Filters. Its application in Biomedical field. Adaptive Filters: Basic Concepts.

Digital signal Processing: Characteristics, frequency domain representation; Stationary and non- stationary bio-signals, waveform detection, Sampling Theory, Finite data considerations (Edge effects), Z Transform, FIR and IIR filters specific to event detection of ECG. Computation of diagnostic signal parameters of ECG like Heart rate and QRS detection using Multivariate analysis like PCA and ICA.

Textbooks/References

1. Joseph J. Carr and John M. Brown, "Introduction to Biomedical Equipment Technology", 4 th Edition, Prentice Hall, 2000.
2. R. Rangayan, "Biomedical Signal Analysis", Wiley 2002.
3. John Semmlow, "Bio-signal and Biomedical Image Processing", Marcel Dekker.
4. R.S.Khandpur, "Handbook of Biomedical Instrumentation", Tata McGraw Hill, New Delhi, 2003, Edition-II.
5. Joseph J. Carr and John M. Brown, "Introduction to Biomedical Equipment Technology", 4 th Edition, Prentice Hall, 2000.
6. Bruce, "Biomedical Signal Processing & Signal Modeling," Wiley, 2001
7. Sörnmo, "Bioelectrical Signal Processing in Cardiac & Neurological Applications", Elsevier.
8. C.Reddy "Biomedical Signal Processing: Principles and techniques", Tata McGraw Hill, New Delhi, 2005.

8. Willis J Tompkins, "Biomedical Signal Processing", ED, Prentice - Hall, 1993.

ECS324 Advanced Digital Signal Processing [3-1-0-4]

Course Pre-requisites

- Digital Signal Processing

Course Objectives

- To develop a comprehensive understanding of the Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) algorithms, including Radix-4 Decimation-in-Time (DIT), Decimation-in-Frequency (DIF), Composite Radix and Split Radix-FFT.
- To explore time-frequency analysis techniques such as Short-Time Fourier Transform (STFT), wavelet transforms, and their applications in multi-resolution signal analysis.
- To understand and implement multi-rate digital signal processing (DSP), including sampling rate conversion, polyphase decomposition, and applications in digital filter banks and subband coding systems.

Course Outcomes

Upon completion of the course, students will be able to:

- Gain a deep understanding of the DFT and FFT, including their implementation using Radix-4 DIT and DIF, Composite Radix and Split Radix-FFT algorithms.
- Use Discrete Cosine Transform (DCT) and Discrete Sine Transform (DST) for various signal processing tasks.
- Perform time-frequency analysis using Short-Time Fourier Transform (STFT) and Wavelet Transform.
- Implement multi-rate digital signal processing techniques, design and implement multistage sampling rate conversion systems.
- Understand and apply digital filter banks and subband coding systems for efficient signal representation and compression.

Syllabus

Overview of Discrete Fourier transform (DFT) and FFT: Radix-4, DIT/DIF FFT algorithms. Composite radix FFT algorithms Split radix-FFT algorithms, Discrete Cosine Transform /Discrete Sin Transform: Need for time frequency analysis, short-time Fourier transform, Multi- resolution concept and analysis, Wavelet transforms.

Introduction to Multi-rate Digital Signal Processing, Sampling, Sampling rate conversion, – Sample rate reduction – decimation by integer factors- sampling rate increase – interpolation by integer factor, Sampling rate conversion by non-integer factor, Multistage Implementation of Sampling Rate Conversion, Filters in Multirate Systems - Filter Design for IIR Interpolators and Decimators, Implementation of Sampling Rate Conversion in FIR Filters, Design of practical sampling rate converters - Filter Specification, filter requirement for individual stages,

Determining the number of stages and decimation factors, Applications of Multirate DSP – Digital Filter Banks - Uniform DFT Filter bank, Nyquist Filters, 2 Channel Quadrature Mirror Filter Bank – perfect reconstruction (PR) or nearly perfect reconstruction (NPR) property, aliasing cancellation condition, perfect reconstruction condition, M Channel Quadrature Mirror Filter Bank - aliasing cancellation condition, perfect reconstruction condition, Subband coding, Subband coding system.

Textbooks/References

1. S. K. Mitra, Digital Signal Processing: A Computer Based Approach. Tata McGraw Hill.
2. John G.Proakis, Dimitris G.Manolakis, Digital Signal Processing, Principles, Algorithms and Applications, Third edition, (2000) PHI.
3. P.P. Vaidyanathan, Multirate systems and filter banks, Prentice Hall, 1993.
4. N.J. Fliege, Multirate digital signal processing- Multirate Systems, Filter Banks, Wavelets, John Wiley 1994.
5. C. K. Chui, An Introduction to Wavelets, Academic Press USA.
6. I. Daubechies, Ten Lectures on Wavelets, SIAM, 1990.
7. Lokenath Debnath, Firdous Ahmad Shah, Wavelet Transforms and Their Applications, Springer, Birkhauser, 2002.
8. S. Mallat, A wavelet Tour of Signal Processing- The sparse way, Academic Press USA 2009. Emmanuel C Ifeachor, Barrie W Jrevis, Digital Signal Processing: a practical approach, Pearson.

ECSXXX ASIC Design [3-1-0-4]

Course Pre-requisites

- Digital System Design with HDL
- VLSI Design

Course Objectives

- To understand full custom and semicustom ASICs
- To learn about ASIC library design
- To know the concept schematic entry hierarchical design and cell library
- To understand programmable ASIC, and simulation examples

Course Outcomes

On completion of this course the student will be able to:

- 1. Describe full custom, semicustom ASICS.
- 2. Explain ASIC library design.
- 3. Describe concept schematic entry hierarchical design and cell library.
- 4. Know the concept of programmable ASIC, and simulation examples.

- 5. Explain ASIC physical synthesis.

Syllabus

Introduction: Full Custom with ASIC, Semi-custom ASICS, Standard Cell based ASIC, Gate array based ASIC, Channeled gate array, Channel less gate array, structured get array, Programmable logic device, FPGA design flow, SIC cell libraries. Data Logic Cells: Data Path Elements, Adders, Multiplier, Arithmetic Operator, I/O cell, Cell Compilers. ASIC Library Design: Logical effort: practicing delay, logical area and logical efficiency logical paths, multi stage cells, optimum delay, optimum number. of stages, library cell design. Low-Level Design Entry: Schematic Entry: Hierarchical design. The cell library, Names, Schematic, Icons & Symbols, Nets, schematic entry, connections, vectored instances and buses, Edit in place attributes, Netlist, screener, Back annotation. Programmable ASIC logic cell, Introduction to Synthesis and Simulation, Examples. Floor Planning and Placement And Routing: Physical Design, CAD Tools, System Partitioning, Estimating ASIC size, partitioning methods. Floor planning tools, I/O and power planning, clock planning, placement algorithms, iterative placement improvement, Time driven placement methods. Physical Design flow global Routing, Local Routing, Detail Routing, Special Routing, Circuit Extraction and DRC.

Recent trends in related areas from Journals like TVLSI, JSSC, TCAS-I etc, Conference proceedings like ISCAS, ISSCC, CICC, etc.

Textbooks/References

1. M.J.S. Smith, - "Application - Specific Integrated Circuits" – Pearson Education, 2003.
2. Neil Weste, K Eshraghian, "Principles of CMOS VLSI design", 4 th Edition, 2010
3. D. A Pucknell "Basic VLSI Design", PHI Publication
4. R. Jacob Baker, Harry W Li, David E Boyce, "CMOS Circuit Design, Layout, Stimulation", PHI Edn, 2005.

ECSXXX VLSI Testing and Verification [3-1-0-4]

Course Pre-requisites

- Digital System Design with Verilog HDL
- VLSI Design

Course Objectives

- To understand the Concepts of VLSI testing and verification techniques.
- To study the concepts of DFT and BIST.
- To understand the concepts of design verification and hardware emulation.
- To study and familiarize EDA verification tools

Course Outcomes

On completion of this course the student will be able to:

- Understand design verification and testing techniques in VLSI circuits.

- Apply testing and verification using EDA tools.

Syllabus

Scope of testing and verification in VLSI design. Introduction to VLSI Design testing, VLSI testing needs and challenges, yield, defects. Fault modeling, Fault equivalence, Fault simulation, Automatic test pattern generation, D Algorithm, PODEM, FAN Design for testability, Scan design, Boundary scan, System test and test for SoCs, Iddq testing, Delay fault testing, BIST for testing logic and memories, Test Automation Design verification techniques based on simulation, analytical and formal approaches, Functional verification, Timing verification, Formal verification, Basics of equivalence checking and model checking, ASIC and FPGA verification, Hardware Emulation Verification Tools from EDA vendors, Sample lab experiments can be given for verification, as exercises

Textbooks/References

1. P. K. Lala, "Digital Circuit Testing and Testability", Academic Press
2. M.L. Bushnell and V.D. Agrawal, "Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits", Kluwer Academic Publishers.
3. M. Abramovici, M.A. Breuer and A.D. Friedman, "Digital Systems and Testable Design", Jaico Publishing House, 2002.
4. Janick Bergeron, "Writing testbenches: functional verification of HDL models", 2nd edition ,Kluwer Academic Publishers, 2003

ICS322 Machine Learning [3-0-0-3]

Course Pre-requisites

- Calculus & Linear Algebra
- Programming & Data Structure Course.

Course Objectives

- To provide an in-depth introduction to supervised, unsupervised and reinforcement learning algorithms.
- To design and implement machine learning solutions to classification, regression, and clustering problems.

Course Outcomes

- Develop an appreciation for what is involved in learning from data.
- Understand a wide variety of learning algorithms.
- Understand how to apply a variety of learning algorithms to data.
- Understand how to perform evaluation of learning algorithms and model selection.

Syllabus

Review of linear algebra, optimization and probability: Matrices, Eigen values and vectors, gradient, hessian, least squares, optimization; random variables and distributions

Definitions, goals and history of Machine Learning; Introduction, linear classification; Classification errors; Regression Techniques.

Supervised learning (generative / discriminative learning, parametric / non-parametric learning, neural networks, support vector machines);

Unsupervised learning (clustering, dimensionality reduction, kernel methods); learning theory (bias/variance trade-offs; VC theory; large margins);

Reinforcement learning and adaptive control. Applications of machine learning.

Textbooks/References

1. Mitchell, Tom. Machine Learning. New York, NY: McGraw-Hill, 1997.
2. Bishop, C. ,M., Pattern Recognition and Machine Learning, Springer, 2006
3. P. Langley, Elements of Machine Learning, Morgan Kaufmann, 1995.
4. Hastie, T., R. Tibshirani, and J. H. Friedman. The Elements of Statistical Learning: Data Mining, Inference and Prediction, Second Edition, Springer, 2009
5. MacKay, David. Information Theory, Inference, and Learning Algorithms. Cambridge, UK: Cambridge University Press, 2003.

ISCXXX Control Systems [3-0-0-3]

Course Objectives

This course will enable students to:

- Understand the basic features, configurations and application of control systems.
- Understand various terminologies and definitions for the control systems.
- Learn how to find a mathematical model of electrical, mechanical and electro- mechanical systems.
- Know how to find time response from the transfer function.
- Find the transfer function via Mason s' rule.
- Analyze the stability of a system from the transfer function.

Course Outcomes

- Develop the mathematical model of mechanical and electrical systems.
- Develop transfer function for a given control system using block diagram reduction techniques and signal flow graph method.
- Determine the time domain specification s for first and second order systems.

- Determine the stability of a system in the time domain using Routh-Hurwitz criterion and Root-locus technique.
- Determine the stability of a system in the frequency domain using Nyquist and bode plots

Syllabus

Introduction to Control Systems: Types of Control Systems, Effect of Feedback Systems, Differential equation of Physical Systems –Mechanical Systems, Electrical Systems, Electromechanical systems, Analogous Systems.

Block diagrams and signal flow graphs: Transfer functions, Block diagram algebra and Signal Flow graphs.

Time Response of feedback control systems: Standard test signals, Unit step response of First and Second order Systems. Time response specifications, Time response specifications of second order systems, steady state errors and error constants.

Introduction to PI, PD and PID Controllers (excluding design). Stability analysis: Concepts of stability, Necessary conditions for Stability, Routh stability criterion, Relative stability analysis: more on the Routh stability criterion.

Introduction to Root-Locus Techniques, The root locus concepts, Construction of rootloci.

Frequency domain analysis and stability: Correlation between time and frequency response, Bode Plots, Experimental determination of transfer function.

Introduction to Polar Plots, (Inverse Polar Plots excluded) Mathematical preliminaries, Nyquist Stability criterion, (Systems with transportation lag excluded)

Introduction to lead, lag and lead- lag compensating networks (excluding design). Introduction to State variable analysis: Concepts of state, state variable and state models for electrical systems, Solution of state equations.

Textbooks/ References

1. "Modern Control Engineering," K.Ogata, Pearson Education Asia/ PHI,4 Edition, 2002. ISBN 978 - 81 - 203 - 4010 - 7.
2. "Automatic Control Systems", Benjamin C. Kuo, John-Wiley India Pvt. Ltd.,8 Edition, 2008.
3. "Feedback and Control System," Joseph J Distefano III et al., Schaum'sOutlines, TMH, 2 nd Edition 2007.

SEMESTER VII

ECE411 FPGA-Based System Design [3-0-2-4]

Course Objectives

- To understand FPGA basics.
- To learn application of FPGA in reconfigurable computing
- To learn case studies.
- To expose students to HDL programming.

Course Outcomes

- Have understood the necessity of reconfigurable architectures.
- Have learned the basic modules of HDL.
- Develop applications using FPGA.

Syllabus

General Purpose Computing Vs Reconfigurable Computing - Simple Programmable Logic Devices - Complex Programmable Logic Devices - FPGAs - Device Architecture - Case Studies.

Compute Models - Programming FPGA Applications in HDL - Compiling C for Spatial Computing - Operating System Support for Reconfigurable Computing.

The Design Flow - Technology Mapping - FPGA Placement and Routing - Configuration Bitstream Generation - Case Studies with Appropriate Tools.

Case Studies of FPGA Applications - System on a Programmable Chip (SoPC) Designs.

Textbooks/References

1. Maya B. Gokhale and Paul S. Graham, "Reconfigurable Computing: Accelerating Computation with Field Programmable Gate Arrays", Springer, 2005.
2. Scott Hauck and Andre Dehon (Eds.), "Reconfigurable Computing - The Theory and Practice of FPGA-Based Computation", Elsevier / Morgan Kaufmann, 2008.
3. Christophe Bobda, "Introduction to Reconfigurable Computing - Architectures, Algorithms and Applications", Springer, 2010.

ECE412 Antenna Theory and Design [3-0-2-4]

Course Objectives

- To understand antenna and radiations.
- To learn application of antennas in microranges.
- To learn how radiation happens through antennas.
- To expose students to newer antenna concepts.

Course Outcomes

- Have understood the necessity of antennas.
- Have learned the basic radiation patterns of antennas.

- Have understood the applications of antennas.

Syllabus

Antenna fundamental parameters, Radiation integrals, Radiation from surface and line current distributions - dipole, monopole, loop antenna; Mobile phone antenna- base station, hand set antenna; Image; Induction, reciprocity theorem, Broad-band antennas and matching techniques, Balance to unbalance transformer, Introduction to numerical techniques.

Field equivalence principle, Radiation from Rectangular and Circular apertures, Uniform aperture distribution on an infinite ground plane; Slot antenna; Horn antenna; Reflector antenna, aperture blockage, and design consideration.

Introduction-General structure of phased array, linear array theory, variation of gain as a function of pointing direction, effects of phase quantization, frequency scanned arrays, analog beam forming matrices-Active modules, digital beam forming, MEMS technology in phased arrays-Retro directive and self phased arrays.

Radiation Mechanism from patch; Excitation techniques; Microstrip dipole; Rectangular patch, Circular patch, and Ring antenna - radiation analysis from transmission line model, cavity model; input impedance of rectangular and circular patch antenna; Microstrip array and feed network; Application of microstrip array antenna. **Lab Practice:**

Simulations and analyses to be carried out using licensed CST Microwave Studio and MATLAB. Characterization, modelling and analysis to be done for the following antennas:

- Dipole and Monopole antenna
- Loop antenna
- Rectangular and pyramidal Horn antenna
- Phased array antenna
- Rectangular and Circular Microstrip Patch antenna
- Ring patch antenna
- Inverted-F antenna
- Microstrip Arrays
- MIMO antennas
- Circular polarization and miniaturization techniques in antennas

Textbooks/References

1. Constantine Balanis, "Antenna Theory Analysis and Design", 4th edition, Wiley, 2016.
2. Hubregt.J.Visser "Antenna Theory and Applications" 1st Edition, John Wiley & Sons Ltd, Newyork, 2012.
3. Zhijun Zhang" Antenna Design for Mobile Devices" 1st Edition, John Wiley & Sons (Asia) Ltd, Newyork,2011.
4. Xavier Begaud, "Ultra Wide Band Antennas" , 1st Edition, ISTE Ltd and John Wiley & Sons Ltd, Newyork,2013.

ECExxx Electronic Measurements and Instrumentation [3-1-0-4]

Course Objectives

- To understand the basics of measurement systems and the errors associated with it.
- To learn techniques involved in the measurement of physical quantities: displacement, force, torque, vibrations, temperature, pressure etc.

Course Outcomes

- Understand the basic concepts, types and methods of measurement
- Understand different types of instruments and their characteristics
- Understand the principle of commonly used instruments.
- Understand the basics for converting a physical parameter into an electrical quantity
- Recognize the principle of measurement of commonly used physical parameters.

Syllabus

Basic concepts of measurement: Significance of measurements, method of measurements, Instruments: type of measuring instruments, classification of measuring instruments, functions of measuring instruments and measurement system, Elements of a generalized measurement system, Performance Characteristics - static characteristics and dynamic characteristics, types of errors in measurement.

Electromechanical instruments: Moving coil instruments, Moving iron instruments, DC Voltmeters, DC Ammeters, Ohmmeters, Megger.

Electronic instruments: Electronic voltmeters, Digital voltmeters, Electronic multimeter (VOM), Q meter.

AC Bridges: DC bridges: Types of bridges Wheatstone's bridge, Kelvin's bridge, AC bridges - Maxwell's bridge, Hay's bridge, Wien's bridge

Transducers: Definition, principle of sensing and transduction, Classification of Transducers, Characteristics of Transducers, Basic requirement of transducers.

Measurement of physical parameters: Displacement measurement - electrical transducers - resistive transducer potentiometer, Inductive transducers - LVDT, Capacitive transducers - using change in area of plates, using change in distance between plates, using change in dielectric constant, pneumatic transducer - flapper-nozzle arrangement, Pressure measurement using - diaphragm, bourdon tube, bellows, manometers, Temperature measurement using - thermistors, thermocouples & RTD. Force measurement: Load Cell - types, basic principle, Strain Gauge - types, basic principle. Torque measurement - strain gauge torque meters, Vibration measurement - seismic transducer piezoelectric accelerometers.

Textbooks/References

1. A. K. Sawhney, "A Course in Electrical and Electronic Measurements and Instrumentation, 19th edition, Dhanpat Rai & Co. Publications.

2. R K Rajput, "Electronic Measurements and Instrumentation", S. Chand, 2009.
3. J.G. Joshi, "Electronic Measurements and Instrumentation", Khanna Publishing House.
4. Uday A. Bakshi, Late Ajay V. Bakshi, "Electronic Measurements and Instrumentation", Technical Publications, 2020.
5. Arun K. Ghosh, Introduction to measurements and Instrumentation, PHI, 4th Edition 2012.
6. A.D. Helfrick and W.D. cooper, Modern Electronic Instrumentation & Measurement Techniques, PHI - 2001.
7. DVS Murthy, Transducers and Instrumentation, PHI 2nd Edition 2013.

ECSXXX Wireless and Cellular Communication [3-1-0-4]

Course Objectives

- Understand the concepts of propagation over wireless channels from a physics standpoint.
- Application of Communication theory both Physical and networking to understand GSM systems that handle mobile telephony.
- Application of Communication theory both Physical and networking to understand CDMA systems that handle mobile telephony.
- Application of Communication theory both Physical and networking to understand LTE- 4G systems.

Course Outcomes

- Explain concepts of propagation mechanisms like Reflection, Diffraction, Scattering in wireless channels.
- Develop a scheme for idle mode, call set up, call progress handling and call tear down in a GSM cellular network.
- Develop a scheme for idle mode, call set up, call progress handling and call tear down in a CDMA cellular network.
- Understand the Basic operations of Air interface in a LTE 4G system.

Syllabus

Mobile Radio Propagation- Large Scale Path Loss - Free Space Propagation Model, Relating Power to Electric Field, Three Basic Propagation Mechanisms-Reflection (Ground Reflection), Diffraction, Scattering, Practical Link Budget. Fading and Multipath-Broadband wireless channel, Delay Spread and Coherence Bandwidth, Doppler Spread and Coherence Time, Angular spread and Coherence Distance. Statistical Channel Model of a Broadband Fading Channel.

The Cellular Concept – Cellular Concept, Analysis of Cellular Systems, Sectoring.

GSM and TDMA Technology: GSM System overview – Introduction, GSM Network and System Architecture, GSM Channel Concept.

GSM System Operations - GSM Identities, System Operations –Traffic cases, GSM Infrastructure Communications (Um Interface).

CDMA Technology: CDMA System Overview – Introduction, CDMA Network and System Architecture.

CDMA Basics – CDMA Channel Concepts, CDMA System (Layer 3) operations, 3G CDMA. LTE – 4G: Key Enablers for LTE 4G – OFDM, SC-FDE, SC-FDMA, Channel Dependant Multiuser.

Resource Scheduling, Multi-Antenna Techniques, Flat IP Architecture, LTE Network Architecture. Multi-Carrier Modulation – Multicarrier concepts, OFDM Basics, OFDM in LTE, Timing and Frequency Synchronization, Peak to Average Ratio, SC-Frequency Domain Equalization, Computational Complexity Advantage of OFDM and SC-FDE.

LTE – 4G: OFDMA and SC-FDMA – Multiple Access for OFDM Systems, OFDMA, SCFDMA, Multiuser diversity and opportunistic scheduling, OFDMA and SC-FDMA in LTE, OFDMA system, Design Considerations.

The LTE Standard – Introduction to LTE and Hierarchical Channel Structure of LTE, Downlink OFDMA Radio Resources, Uplink SC-FDMA Radio Resources.

Textbooks/References

1. Fundamentals of LTE” Arunabha Ghosh, Jan Zhang, Jeffrey Andrews, Riaz Mohammed, Pearson education (Formerly Prentice Hall, Communications Engg and Emerging Technologies), ISBN-13: 978-0-13-703311-9.
2. “Introduction to Wireless Telecommunications Systems and Networks”, Gary Mullet, First Edition, Cengage Learning India Pvt Ltd., 2006, ISBN - 13: 978-81-315-0559-5.
3. “Wireless Communications: Principles and Practice” Theodore Rappaport, 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2002, ISBN 0-13-042232-0.
4. LTE for UMTS Evolution to LTE-Advanced’ Harri Holma and Antti Toskala, Second Edition -2011, John Wiley & Sons, Ltd. Print ISBN: 9780470660003.

ECSXXX Statistical Signal Processing [3-1-0-4]

Course Prerequisites

- Basic knowledge in probability
- matrices and Digital Signal Processing

Course Objectives

- To develop an understanding of statistical signal processing techniques for analyzing discrete-time random signals and processes.
- To provide the knowledge and skills for energy and power spectrum estimation, using both non-parametric and parametric methods.

- To introduce the concepts and implementation of adaptive filtering algorithms, with a focus on LMS and RLS methods.
- To teach the fundamentals and applications of Kalman filtering for optimal state estimation in dynamic systems.

Course Outcomes

- Understand the fundamental concepts of random variables, random processes, and statistical signal processing.
- Be proficient in techniques for energy and power spectrum estimation using both non-parametric and parametric methods.
- Gain practical experience in the design and implementation of adaptive filters, including LMS and RLS algorithms.
- Apply Kalman filtering techniques for optimal state estimation in discrete and continuous systems.
- Analyze the performance of various statistical signal processing methods through real-world applications like noise cancellation, system identification, and equalization.

Syllabus

Discrete-Time Random Signal Processing – Introduction, Random Variables, Ensemble Averages of a Random Variable, Two-Dimensional Random Variables, Bias and Consistency for Parameter Estimation, Random Processes, Stationary Process, Autocorrelation and Autocovariance Matrices, Ergodic Processes, Power Spectrum, Filtering Random Processes, Spectral Factorisation Theorem. Energy and Power Spectrum Estimation - Introduction, Energy Spectrum of Discrete Time Signal, Power Spectrum of a Random Process, Energy Spectral Density, Estimation of the Autocorrelation and Power Spectrum of Random Signals, Periodogram, Use of DFT/FFT in Power Spectrum Estimation, Power Spectrum Estimation: Non-Parametric Methods - Bartlett method, Welch method, Blackman-Tukey method, AR, MA and ARMA Models, Power Spectrum Estimation: Parametric Methods - AR Model (Yule-Walker Method), The Levinson-Durbin Recursion, Optimum Filters. Adaptive Filtering: Concept of adaptive filtering, Basic Wiener filter theory, Wiener Hoff Equation, Steepest Descent Algorithm (SDA), Least mean square (LMS) adaptive algorithm - Implementation of LMS adaptive algorithm, Practical limitations of LMS adaptive algorithm, Recursive Least Square (RLS) algorithm - Limitations of RLS adaptive algorithm, Applications of adaptive filters – System Identification, Noise cancellation, Echo cancellation, Equalization. Kalman filtering: State-space model and the optimal state estimation problem, discrete Kalman filter, continuous-time Kalman filter, extended Kalman filter.

Textbooks/References

1. D.H. Johnson. Statistical Signal Processing. Available as a coursepak at the Rice Campus Store.
2. Monson H. Hayes, Statistical Digital Signal Processing and Modeling, John Wiley & Sons, Inc, Edition 2002.

3. S. M. Kay, Fundamentals of Statistical Signal Processing, Vol I: Estimation Theory, Vol II: Detection Theory, Prentice Hall, 1993/1998.
4. Charles W. Therrien, Discrete Random Signals and Statistical Signal Processing, Prentice Hall Signal Processing Series, Edition 2004.
5. D.G. Manolakis, V.K. Ingle and S.M. Kogon, Statistical and Adaptive Signal Processing, McGraw Hill, Edition 2000.
6. Gray, R. M. and Davisson L. D., An Introduction to Statistical Signal Processing, Cambridge University Press, 2004 (<http://www.ee.stanford.edu/~gray/sp.html>).

ECSXXX Analog IC Design [3-1-0-4]

Course Prerequisites

Analog Electronics

Course Objectives

The course is aimed to

- Analyze and design of operational amplifier
- Study of effect of noise in analog circuits
- Emphasize the design of practical amplifiers, and their performance parameters tradeoffs.
- Study the design of oscillator .
- Study the basics of biasing schemes in Analog IC design.

Course Outcomes

At the end of the course the student will be able to

- Understand the design challenges in nanometer design.
- Design multistage operational amplifier
- Understand frequency compensation techniques of op-amp and its stability analysis
- Understand the fundamentals of bandgap reference circuit.

Syllabus

Nanometer design studies- Design methodology of a basic MOSFET amplifier with biasing- Determining voltage swing, and parameters affecting gain of Amplifier. Design of five transistor operational transconductance amplifier. Frequency response of differential amplifier. Noise- Noise Analysis in basic MOSFET Amplifier topologies, Noise in differential amplifier. Operational amplifiers- Need for Single and Multistage amplifiers, Design of cascode, folded cascode and two-stage op-amps. Slewrate in two two stage op-amp. Gain boosting technique in opamp design. Stability and frequency compensation- Basic Concepts, Instability condition, Effect of Pole Locations on Stability, Frequency Compensation in operational amplifiers- Concepts and Techniques for Frequency Compensation – Dominant pole, Miller Compensation in two stage op-amp., Compensation of Miller RHP Zero. Oscillator- Design issues, Need of negative resistance in oscillator designs. Introduction to Biasing Circuits- Bandgap Reference circuit.

Textbooks/References

- Behzad Razavi, Design of Analog CMOS Integrated Circuits, 2017, Second Edition, McGraw-Hill.
- Phillip E. Allen and Douglas R. Holberg, CMOS Analog Circuit Design, 2016, Second Edition, Oxford University Press.
- David Johns and Ken Martin, Analog Integrated Circuit Design, 2012, Second Edition, John Wiley & Sons, Inc.
- Gray, P. R., Hurst, P. J., Lewis, S. H., & Meyer, 2009 Analysis and Design of Analog Integrated Circuits, 5th Edition John Wiley & Sons.

ECSXXX Nanoelectronics [3-1-0-4]

Course Objectives

The course intends to give students a broad understanding of fundamentals, fabrication technologies and applications of nanoscale structures. Students will also be trained for literature study and critique, oral presentation, problem formulation, solution development, and formal writing.

Course Outcomes

The student will be able to explain the significance of current research about a particular topic in nanoelectronics.

Syllabus

Introduction (from classical electronics to nanoelectronics) Wave-particle duality, Schrödinger wave equation, Materials for nanoelectronics - Semiconductors - Carbon nanomaterials nanotubes.

Electrons in low-dimensional structures Electrons in quantum wells - Electrons in quantum wires - Electrons in quantum dots, Fabrication of nanostructures - Crystal growth - Nanolithography

Clusters and nanocrystals - Nanotube growth Characterization of nanostructures

Electron transport in semiconductors and nanostructures - Time and length scales of the electrons

in solids - Statistics of the electrons in solids and low dimensional structures - Electron transport in nanostructures

Nanoelectronic devices - Resonant-tunneling diodes - Field-effect transistors - Single-electrontransfer devices.

Textbooks/References

1. Vladimir V. Mitin, Viatcheslav A. Kochelap, Michael A. Strosio, "Introduction to Nanoelectronics", Cambridge University Press, 2008. (ISBN: 9781107403765)
2. George W. Hanson, "Fundamentals of nanoelectronics", Pearson/Prentice, 2008. Quantum Mechanics
3. D. J. Griffiths, "Introduction to quantum mechanics (2nd edition)", Prentice Hall, 2005. Solid- state physics
4. R. Turton, "The physics of solids", Oxford, 2000. Semiconductor physics and devices
5. S. M. Sze, "Physics of Semiconductor Devices (2nd)", Wiley, 1981
6. S. Datta, "Lessons from Nanoelectronics: A New Perspective on Transport (Lessons from Nanoscience: a Lecture Notes Series) World Scientific, 2012

7. V. Mitin, V. Kochelap, and M. Strosio "Introduction to Nanoelectronics: Science, Nanotechnology, Engineering, and Applications", Cambridge University Press, 2008.
8. C. P. Poole and F. J. Owens, "Introduction to nanotechnology", John Wiley & Sons, 2003

IOEXXX Cryptography and Network Security [3-0-0-3]

Course Objectives

- To lay a foundation on Security in Networks, attacks, defence and Classical Cryptosystems
- To analyse various Private and Public key Cryptosystems to ensure confidentiality, Integrity and Authentication.
- To analyse various protocols to ensure Email Security and Network Security.
- To apply Cryptography in various Applications.

Course Outcomes

- Understand the fundamental concepts of Cryptography, Types of Security breaches, attacks, defence, control measures, Classical Cryptosystem.
- Compare various Private and Public key Cryptosystems to ensure confidentiality, Integrity and Authentication.
- Understand various protocols in Email Security and Network Security.
- Apply Cryptography in various Applications.

Syllabus

Introduction to Security in networks, Types of Security breaches, attacks, defence, control measures, Classifying cryptosystems, classical cryptosystems, block cipher modes of operation, DES encryption and decryption, triple DES, AES encryption and decryption.

Public Key Cryptosystem - RSA cryptosystem, Diffie-Hellman Key Exchange Algorithm, Elliptic curve cryptosystem, Message Authentication and Hash Function- MD5 message digest algorithm, Secure hash algorithm, Authentication Protocols and Digital signature, DSS.

Kerberos - X. 509 Authentication Service. - Pretty Good Privacy - Electronic Mail Security - IP Security Architecture - Web Security Considerations - Secure Socket Layer and Transport Layer Security - Secure Electronic Transaction - Firewalls - Firewall Design Principles

Applications of Cryptography- Blockchain, Bitcoin and Cryptocurrency Technologies

Textbooks/References

1. Behrouz A. Forouzan, Debdeep Mukhopadhyay, Cryptography and Network Security, 3rd Edition, Mc Graw Hill Education, 2016.
2. Stallings W., Cryptography and Network security: Principles and Practice, 7/e, Pearson Education Asia, 2017.

3. Charles P. Pleege, Shari Lawrence Pleege, "Security in Computing", Pearson Education Asia, 4th Edition, 2009.
4. Alan T. Norman, "Blockchain Technology Explained: The Ultimate Beginner's Guide About Blockchain Wallet, Mining, Bitcoin, Ethereum, Litecoin, Zcash, Monero, Ripple, Dash, IOTA and Smart Contracts" Kindle Edition, 2017.
5. Andreas Antonopoulos, "Mastering Bitcoin: Unlocking Digital Cryptocurrencies" 1st Edition, 2014.
6. Matthew Connor, "Blockchain: Ultimate Beginner's Guide to Blockchain Technology Cryptocurrency, Smart Contracts, Distributed Ledger, Fintech, and Decentralized Applications" Kindle Edition, 2017.

IOEXXX Digital Image Processing [3-0-0-3]

Course Objectives

- To study the image fundamentals and selected mathematical transforms for image processing.
- To apply image processing techniques for image enhancement, restoration, segmentation and recognition.
- To provide understanding about image compression techniques and standards.

Course Outcomes

- Learn image representation and transformation in digital domain.
- Implement techniques for image enhancement, restoration, segmentation and recognition.
- Assess image compression methods and standards.

Syllabus

Digital Image Fundamentals: Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels - Colour image fundamentals - RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms - DFT, DCT

Image Enhancement: Spatial domain methods: point processing- intensity transformations, histogram processing, image subtraction, image averaging. Spatial filtering- smoothing filters, sharpening filters. Frequency domain methods- low pass filtering, high pass filtering, homomorphic filtering Image Restoration: Degradation model, Linear Filtering- Inverse and Wiener, Nonlinear filtering- Mean, Median, Max and Min filters, Geometric transformations-spatial transformations

Image Segmentation: Edge detection, Edge linking via Hough transform – Thresholding - Region based segmentation – Region growing – Region splitting and merging – Morphological processing- erosion and dilation, Segmentation by morphological operations

Image Compression and Recognition: Need for data compression, Huffman, Run Length Encoding, Arithmetic coding, JPEG standard, Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors– Topological feature, Texture - Patterns and Pattern classes – Image Recognition.

Textbooks/References

1. Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, Digital Image Processing using MATLAB[®], Pearson Education, Inc., 2011
2. S Jayaraman, S Esakkirajan, T Veerakumar, Digital image processing, Tata Mc Graw Hill, 2015
3. Kenneth R Castleman , Digital image processing:, Pearson Education, 2/e, 2003
4. Pratt William K , Digital Image Processing: , John Wiley, 4/e, 2007
5. Jain Anil K , Fundamentals of digital image processing, PHI, 1988

IOEXXX Pattern Recognition [3-0-0-3]

Course Objectives

This course covers the techniques and gain proficiency of pattern recognition that are fundamental to a wide variety of application areas such as medical research, biometrics, computer vision, etc.

Course Outcomes

- A good knowledge of Bayesian decision theory and Bayesian learning.
- Fundamental understanding of classifiers such as linear discriminant function, quadratic discriminant function, nearest neighbor rule, neural network and SVM.
- A good understanding of feature selection algorithms.
- Ability to evaluate the performance of various classifiers on real-world datasets.

Syllabus

1) Basics of Probability, Random Processes and Linear Algebra (recap): Probability: independence of events, conditional and joint probability, Bayes theorem Random Processes: Stationary and non-stationary processes, Expectation, Autocorrelation, Cross-Correlation, spectra. 2) Bayes Decision Theory: Minimum-error-rate classification. Classifiers, Discriminant functions, Decision surfaces. Normal density and discriminant functions. Discrete features. 3) Parameter Estimation Methods: Maximum-Likelihood estimation: Gaussian case. Maximum Posteriori estimation. Bayesian estimation: Gaussian case. Unsupervised learning and clustering - Criterion functions for clustering. Algorithms for clustering: K-Means, Hierarchical and other methods. Cluster validation. Gaussian mixture models, Expectation-Maximization method for parameter estimation. Maximum entropy estimation. 4) Sequential Pattern Recognition. Hidden Markov Models (HMMs). Discrete HMMs. Continuous HMMs. Nonparametric techniques for density estimation. Parzen-window method. K-Nearest Neighbour method. 4) Dimensionality reduction: Principal component analysis - it relationship to Eigen analysis. Fisher discriminant analysis - Generalized Eigen analysis. Eigen vectors/Singular vectors as dictionaries. Factor Analysis, Total variability space - a dictionary learning methods. Non negative matrix factorization - a dictionary learning method. 5) Linear discriminant functions: Gradient descent procedures, Perceptron, Support vector

machines - a brief introduction. 6) Artificial neural networks: Multilayer perceptron – feed forward neural network. A brief introduction to deep neural networks, convolutional neural networks, recurrent neural networks.(4) 7) Non-metric methods for pattern classification: Non-numeric data or nominal data. Decision trees: Classification and Regression Trees (CART).

Textbooks/References

1. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001
2. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009
3. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006
4. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006
5. P.A Devijver and J. Kittler, Pattern Recognition: A Statistical Approach, Prentice-Hall International, Englewood Cliffs, NJ, 1980.
6. K. Fukunaga, Introduction to Statistical Pattern Recognition, 2nd Ed. Academic Press, NewYork, 1990.

SEMESTER VIII

ECSXXX Real Time Embedded Systems [3-0-0-3]

Course Objectives

- To understand processors and their instruction sets .
- To understand hardware platform for embedded systems.
- To design and analysis programs for embedded systems.
- To understand distributed and multiprocessor embedded systems.

Course Outcomes

- Have understood the approach to develop assembly code for processors.
- Have learned to perform platform level performance analysis.
- Have understood develop applications using RTOS.

Syllabus

Introduction to embedded computing - overview of embedded system design process - instruction sets of processors: ARM, PIC, TI C55x, TI C64x - programming I/O - modes and exceptions coprocessors - memory system - CPU performance-CPU power consumption

Basic computing platforms - CPU Bus - memory devices and systems - choosing a platform development environments - debugging consumer electronics architecture - platform-level performance analysis - design example: Audio Player

Components for embedded programs - models of programs - Assembly, linking, and loading compiler optimizations -

program-level performance analysis - performance optimization
- program-level energy optimization - optimizing program size
- program validation and testing design example: Digital Still Camera

System design methodologies - requirements analysis - specifications - architecture design quality assurance - distributed embedded systems

shared-memory multiprocessors - design example: Video accelerator

Textbooks/References

1. Christopher Hallinan, "Embedded Linux Primer: A Practical Real-World Approach", Second Edition, Prentice Hall, 2010.
2. Karim Yaghmour et al., "Building Embedded Linux Systems", O'Reilly, 2008 .
3. Arnold S. Berger, "Embedded Systems Design: An Introduction to Processes, Tools, and Techniques", CMP Books, 2001.
4. David E. Simon, "An embedded Software Primer", Addison-Wesley, 1999.

ECSXXX Optoelectronics [3-0-0-3]

Course Prerequisites// Electronic Devices or equivalent. Course also re-visits the basic device fundamentals required for understanding optoelectronic devices earlier in the course.

Course Objectives

- This course provides the background in optoelectronics, help students meet the demand of growing semiconductor optoelectronic industry and prepares them to advanced study and research in the semiconductor optics and optoelectronics devices.
- The course is an introduction to the fundamentals of optoelectronics and principles of the optoelectronic devices operation.
- Topics include a variety of different subjects including a detailed discussion of the design and operation of optical LEDs, the basic physics and operation of lasers and photodetectors
- Familiarize with the most promising silicon optical components and the path to electronic-photonics integration.
- Emphasis is on the underlying device physics behind the operation and design of optoelectronic devices.
- Identify current research themes and technologies in the field of optoelectronic devices and integration, study the relevant literatures and present a critical analysis of the results.

Course Outcomes Upon Completion of the course, the students will be able to

- Be familiar with recent trends in optoelectronics.
- Understand the electromagnetic spectrum, wave equation and wave propagation in linear, isotropic media and in anisotropic media.

- Understand basic principles of light propagation and modal analyses of optical fiber. Understand the basic operating principles of light sources, detectors.
- Identify trends in optical interconnection and the power of electronic-photonics convergence
- Research-led teaching will be included in this course through lectures on review of related research field and progress, and group assignments on various research topics.
- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Syllabus

Electromagnetic theory: Maxwell's equations, Poynting's theorem, Wave nature of light, Polarization, Interference, Diffraction, Light Source, review of Quantum Mechanical concept, Review of Solid State Physics, Review of Semiconductor Physics and Semiconductor Junction Device.

FIBRE OPTICS AND APPLICATIONS Fibre Bragg Grating, Long- period fibre Bragg Grating. Fabrication of Fibre Gratings. Michelson's interferometer, Mach-Zehnder interferometer, Optical Fibre sensors: Intensity modulation sensors. Phase modulation sensors, Temperature, pressure, chemical and rotation sensors. Fibre optic gyroscopes. Evanescent wave sensors, Optical Biosensors.

SILICON-ON-INSULATOR (SOI): PHOTONICS, Integration of photonic devices, power and power density issues on-chip, Advanced waveguides and devices, Testing and packaging of Silicon photonic devices: Silicon Photonics Research and Manufacturing using SOI wafer. Silicon Photonics Application. Optoelectronic devices, Optical modulators, modulation methods and modulators, transmitters, optical transmitter circuits, LED and laser drive circuits, LED-Power and efficiency, double heterostructure LED, LED structures, LED characteristics, laser modes, strip geometry, gain guided lasers, index guided lasers. Display devices, Photo detectors, photon devices, PMT, photodiodes, photo transistors, noise characteristics, PIN diode, APD characteristics, APD Design of detector arrays, CCD, Solar cells.

OPTICAL COMMUNICATION TECHNOLOGY: Un-guided optical communication system, transmission parameters, beam divergence, atmospheric attenuation, guided wave communication, merits of optical fibre communication systems, basic network information rates, time evolution of fibre optic systems, elements of optical fiber transmission link/ repeaters, integrated optics, active and passive components, opto-mechanical switches, all optical switches.

LASER TECHNOLOGY: Laser Emission, Absorption, Radiation, Population Inversion, Optical Feedback, Threshold condition, Laser Modes, Radiative decay of excited states of atoms, spontaneous emission, Planck's law for cavity radiation, Q switching general theory, Mode locking, Laser systems, Classes of Lasers.

LASER APPLICATIONS Industrial applications: Absorption of laser radiation by metals, semiconductors and insulators, laser drilling, welding, cutting and surface cleaning, optical fibre splicing, laser deposition of thin films.

LASERS IN MEDICINE: Optical coherence Tomography, Photodynamic therapy, laser angioplasty, Lasers in surgery.

OTHER APPLICATIONS OF LASERS: Laser pollution monitoring, LIDAR laser gyros, Laser spectroscopy, laser induced fusion reactor, Holography and speckle interferometry

Textbooks/References

1. Fundamentals of Photonics - Saleh and Teich Wiley Intsc (2007)
2. Opto electronics - An introduction - J Wilson and J F B J iS Hawkers (Prentics-Hall India, 1996)
3. J. Singh, Optoelectronics: an Introduction to Materials and Devices , McGraw Hill 1996.
4. Silicon Photonics: Fueling the Next Information Revolution by Daryl Inniss, Roy Rubenstein
5. Introduction to fiber optics - Ajoy Ghatak and K. Thyagarajan, Cambridge Univ Press, 1999
6. Laser Fundamentals - Willaim T Selfvast, Cambridge Univ-Press (1996). (Text)
7. Optical Fibre Communication –G Keiser (3rd Ed), 2000

ECSXXX Mixed Signal Design [3-0-0-3]

Course Prerequisites

- VLSI Design
- Analog IC Design

Course Objectives

The course is aimed to

- • Analyze Mixed signal Design PLL and data converters
- • Study of sample and hold circuits
- • Study of switched capacitor circuits for Aanlog to digital Converters
- • Study the basics of biasing schemes in Analog IC design.

Course Outcomes

At the end of the course the student will be able to

- • Design issues in Phase locked loop.
- • Design Sample and Hold Circuits using MOS by considering the non-idealities.
- • Analyze CMOS based Switched Capacitor Circuits.
- • Understand basics of Data Converters.
- • Analyze the architectures of ADCs and DAC.

Syllabus

PLL Design –Simple PLL, Charge pump PLL, Non ideal effects in PLL, Delay locked loop. Sampling circuits- Distortion due to switch - Charge injection - Thermal noise in sample and Hold circuits. Switched Capacitor (SC) circuits– Parasitic Insensitive Switched Capacitor amplifiers. Data converter fundamentals:

Offset and gain Error - Linearity errors – Dynamic Characteristics - Quantization noise spectrum. Introduction to Analog to Digital Converter Architectures, Digital to Analog Converter Architectures

Textbooks/References

1. 1. Behzad Razavi, Design of Analog CMOS Integrated Circuits, 2017, Second Edition, McGraw-Hill.
2. 2. Frank Ohnhaus, Analog-Digital Converters for Industrial Applications Including an Introduction to Digital-Analog Converters, 2015, Springer
3. 1. David Johns and Ken Martin, Analog Integrated Circuit Design, 2012, Second Edition, John Wiley & Sons, Inc.
4. 2. Ahmed M.A.Ali, High Speed Data Converters, 2016, First Edition, IET Materials, Circuits & Devices

ECSXXX Nanoelectronics [3-1-0-4]

Course Objectives

The course intends to give students a broad understanding of fundamentals, fabrication technologies and applications of nanoscale structures. Students will also be trained for literature study and critique, oral presentation, problem formulation, solution development, and formal writing.

Course Outcomes

The student will be able to explain the significance of current research about a particular topic in nanoelectronics.

Syllabus

Introduction (from classical electronics to nanoelectronics) Wave-particle duality, Schrödinger wave equation, Materials for nanoelectronics - Semiconductors - Carbon nanomaterials nanotubes.

Electrons in low-dimensional structures Electrons in quantum wells - Electrons in quantum wires - Electrons in quantum dots, Fabrication of nanostructures - Crystal growth - Nanolithography

Clusters and nanocrystals - Nanotube growth Characterization of nanostructures

Electron transport in semiconductors and nanostructures - Time and length scales of the electrons

in solids - Statistics of the electrons in solids and low dimensional structures - Electron transport in nanostructures

Nanoelectronic devices - Resonant-tunneling diodes - Field-effect transistors - Single-electrontransfer devices.

Textbooks/References

1. Vladimir V. Mitin, Viatcheslav A. Kochelap, Michael A. Strosio, "Introduction to Nanoelectronics", Cambridge University Press, 2008. (ISBN: 9781107403765)
2. George W. Hanson, "Fundamentals of nanoelectronics", Pearson/Prentice, 2008. Quantum Mechanics
3. D. J. Griffiths, "Introduction to quantum mechanics (2nd edition)", Prentice Hall, 2005. Solid- state physics
4. R. Turton, "The physics of solids", Oxford, 2000. Semiconductor physics and devices

5. S. M. Sze, "Physics of Semiconductor Devices (2nd)", Wiley, 1981
6. S. Datta, "Lessons from Nanoelectronics: A New Perspective on Transport (Lessons from Nanoscience: a Lecture Notes Series) World Scientific, 2012
7. V. Mitin, V. Kochelap, and M. Strosio "Introduction to Nanoelectronics: Science, Nanotechnology, Engineering, and Applications", Cambridge University Press, 2008.
8. C. P. Poole and F. J. Owens, "Introduction to nanotechnology", John Wiley & Sons, 2003
3. K R Rao, Zoran S, Bojkovic and Dragorad A, Milovanovic "Multimedia communication systems", PHI, 2003.
4. Jon Crowcroft, Mark Handley, Ian Wakeman "Internet-working Multimedia" Harcourt, Singapore, 1998.
5. Tay Vaughan, "Multimedia Making it to work", 4th edition Tata McGraw Hill, New Delhi, 2000.

ECSXXX Multimedia Communication [3-1-0-4]

Course Objectives

- To understand multimedia networking .
- To understand reliable transport protocol and applications.
- To understand broadband network technology.
- To understand a few multimedia protocols.

Course Outcomes

- Have understood the applications of multimedia networking.
- Have learned about distributed virtual reality.
- Have understood multimedia compression techniques.

Syllabus

Digital Sound, Video and Graphics - Basic Multimedia Networking - Multimedia Characteristics - Evolution of Internet Services Model - Network Requirements for Audio/ Video Transform - Multimedia Coding and Compression for Text, Image Audio And Video.

Broadband Services - ATM and IP, IPV6, High Speed Switching - Resource Reservation, Buffer Management - Traffic Shaping - Caching Scheduling and Policing, Throughput, Delay and Jitter Performance - Storage and Media Services Voice and Video Over IP - MPEG-2 over ATM/IP - Indexing Synchronization of Requests - Recording and Remote Control .

Multicast over Shared Media Network - Multicast Routing and Addressing - Scaling Multicast and NBMA Networks - Reliable Transport Protocols - TCP Adaptation Algorithm - RTP, RTCP MIME. Peer-to-Peer Computing - Shared Application - Video Conferencing, Centralized and Distributed Conference Control - Distributed Virtual Reality - Light Weight Session Philosophy.

Textbooks/References

1. Ivan Vidal, Ignacio Soto, Albert Banchs, Jaime Garcia, Ivan Lozano, Gonzalo Camarillo, Multimedia Networking Technologies, Protocols, Architectures, in Artech Publishers, 2019.
2. B O Szuprowicz, "Multimedia Networking", McGraw Hill, Newyork, 1995.

ECSXXX Robotics and Automation [3-1-0-4]

Course Objectives

- To introduce the basic concepts of robotics and its specifications.
- To introduce the different sensors and actuators used in robots.
- To introduce the various end effector configuration and their design considerations for a particular application.
- To impart knowledge of the basic kinematic model of robotic manipulators.
- To introduce the basics of control of robotic manipulators

Course Outcomes

- Describe the anatomy and specifications of Robots.
- Summarize different sensors and actuators for robots.
- Choose a robotic configuration and grippers for a particular application.
- Obtain basic kinematic model of robotic manipulators.
- Understand the basics of control of robotic manipulators

Syllabus

Introduction: Definition and origin of robotics, Robot Anatomy, Robot specifications, Robot characteristics - accuracy, precision, and repeatability, Classification of robots. Anatomy of a robotic manipulator- links, joints, actuators, sensors, end effector, controller.

Sensors in robotics: Sensor classification: Internal sensors- Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors- Range, proximity, touch, force-torque sensing.

Actuators for robots: Actuators classification Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; End effectors classification, Mechanical grippers, Magnetic grippers, Vacuum grippers, Adhesive grippers, selection and design considerations of grippers in robot.

Robotic vision systems: Imaging, Sensing and Digitization, Image processing techniques.

Robot Kinematics: Homogenous coordinates and transfer representations. Kinematics chains. Introduction to forward kinematics and inverse kinematics.

Robot Programming: Programming methods, Robot language classification, Robot language structure, elements and its functions. Motion, Endeffector and Sensor commands in VAL programming language.

Textbooks/References

1. Mikell P. Groover et. al., "Industrial Robots Technology, Programming and Applications", McGraw Hill, New York, 2008.
2. Saeed B. Niku, "Introduction to Robotics", Pearson, 2011.
3. S K Saha "Introduction to Robotics", Mc Graw Hill Education.
4. John J.Craig, "Introduction to Robotics", Pearson, 2009.
5. R K Mittal and I J Nagrath, "Robotics and Control", Tata McGraw Hill, New Delhi, 2003.
6. Spong & Vidyasagar, Robot Dynamics and Control, McGraw Hill.
7. Fu K S, Gonzalez R C, Lee C.S.G, "Robotics : Control, Sensing, Vision and Intelligence", McGraw Hill, 1987

ECSXXX Advanced Wireless Communication Techniques [3-1-0-4]

Course Objectives

- To extend the knowledge of digital communication under fading channels
- To study the various communication strategies adapted in MIMO system to increase data rate and reliability of communication systems
- To study the theoretical BER performance of various communication systems
- Understand various advanced communication techniques that are proposed for future communication systems.

Course Outcomes

- Knowledge on the performance of digital communication systems under fading channels
- Calculate the BER performance of various communication systems
- Applying various strategies to improve the BER and data rate of communication systems
- Knowledge on various advanced communication techniques

Syllabus

SISO: System model – SISO channel – Channel capacity - BER performance of wireless communication systems – Spatial Diversity: Diversity gain – Coding gain – Spatial diversity – Time/frequency diversity – Transmit antenna diversity – Receive antenna diversity – SIMO: System model – SIMO channel – Channel capacity - Diversity combining techniques - MIMO: System model – MIMO channel – MIMO channel capacity – Spatial Multiplexing – Receiver design: Sphere decoding – V-BLAST – D-BLAST – MIMO detection – Optimum detection – linear detection – Space time block codes – Alamouti Scheme. Cooperative Communications: System model – Cooperation protocols – Amplify and forward (AF) – Decode and forward (DF) – Cooperative communication with single branch: BER performance

of AF and DF protocols – Multi-node cooperative communication – Relay selection – Differential modulation for cooperative communication. Definition of Massive MIMO, Correlated Rayleigh Fading, System Model for Uplink and Downlink, Basic Impact of Spatial Channel Correlation, Channel Hardening and Favourable Propagation, Local Scattering Spatial Correlation Model, Massive MIMO propagation channel, Favorable Propagation and Deterministic Channels, Favorable Propagation and Random Channels-Independent Rayleigh Fading-Uniformly Random Line-of-Sight (UR-LoS), Uplink Pilots and Channel Estimation - Orthogonal Pilots- De-Spreading of the Received Pilot Signal-MMSE Channel Estimation, Uplink Data Transmission - Zero-Forcing -Maximum-Ratio, Downlink Data Transmission-Linear Precoding-Zero-Forcing-Maximum-Ratio. Index Modulation: Space shift keying – Spatial modulation – Media based modulation – Non-Orthogonal Multiple Access (NOMA) – Simultaneous Wireless Information and Power Transfer (SWIPT) – Reconfigurable intelligent surfaces for wireless communications.

Textbooks/References

1. Chockalingam and B. Sundarajan, "Large MIMO Systems", Cambridge University Press, 2014.
2. E. Biglieri, R. Calderbank, A. Constantinides, A. Goldsmith, A. Paulraj and H. V. Poor, "MIMO Wireless Communications", Cambridge University Press, 2007.
3. A. Goldsmith, "Wireless Communications" Cambridge University Press, 2005.
4. K. J. Ray, Ahmed K Sadek, Weifung Su, and Andres Kwasinski "Cooperative Communications and Networking". Cambridge University Press, 2009.
5. T. S. Rappaport, "Wireless Communications: Principles and Practices", Pearson Education, 2002.
6. D. Tse and P. Viswanath, "Fundamentals of Wireless Communication", Cambridge University Press, 2005.
7. Thomas L. Marzetta, Erik G. Larsson, Hong Yang, Hien Quoc Ngo, "Fundamentals of Massive MIMO", Cambridge University Press 2016.
8. Emil Björnson, Jakob Hoydis and Luca Sanguinetti, "Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency", Foundations and Trends, Now, 2017.
9. Miaowen Wen, Xiang Cheng, Liuqing Yang, Index Modulation for 5G Wireless Communications, Springer, 2017.
10. Yuanwei Liu, Zhijin Qin, Zhiguo Ding, Non-Orthogonal Multiple Access for Massive Connectivity, Springer, 2020.
11. Hongliang Zhang, Boya Di, Lingyang Song, Zhu Han, Reconfigurable Intelligent Surface-Empowered 6G, Springer, 2021.

ECSXXX Speech and Audio Processing [3-1-0-4]

Course Prerequisites

Signals and Systems

Course Objectives

- To familiarize basic mechanism of speech production and methods for speech analysis.
- To give an overall picture about various applications of speech processing
- To impart ideas of perception of sound, psycho-acoustic analysis, spatial audio perception and rendering.
- To analyse audio compression schemes and audio quality

Course Outcomes

- Understanding of basic mechanism of speech production and methods for speech analysis
- Develop systems for various applications of speech processing
- Understanding about human hearing system, psycho-acoustic analysis and spatial sound
- Knowledge about audio compression methods and audio quality analysis

Syllabus

Speech Production: Acoustic theory of speech production. Speech Analysis: Speech signal, Short-Time Speech Analysis, Time domain analysis (Short time energy, short time zero crossing Rate, ACF). LPC Analysis (LPC model, Auto correlation method).

Frequency domain analysis (Filter Banks, STFT, Spectrogram), Cepstral Analysis, MFCC. Probabilistic formulation of speech recognition, Speech coding: fundamentals, Comparison of waveform coding, vocoding and hybrid coding, Speech enhancement: fundamentals, basic types, Speaker verification (block diagram) Signal Processing Models of Audio Perception: Basic anatomy of hearing System. Auditory Filter Banks, Psycho-acoustic analysis: Critical Band Structure, Absolute Threshold of Hearing, Simultaneous Masking, Temporal Masking, MPEG psycho-acoustic model. Audio compression methods: Sampling rate and bandwidth requirement for digital audio, Redundancy removal and perceptual irrelevancy removal, Transform coding of digital audio: MPEG2-AAC coding standard, MDCT and its properties, Loss less coding methods.

Spatial Audio Perception and rendering: The physical and psycho-acoustical basis of sound localization and space perception. Spatial audio standards. Audio quality analysis: Objective analysis methods- PEAQ, Subjective analysis methods - MOS score, MUSHRA score

Textbooks/References

1. Douglas O'Shaughnessy, Speech Communications: Human & Machine, IEEE Press, Hardcover 2/e, 1999; ISBN: 0780334493.
2. Nelson Morgan and Ben Gold, Speech and Audio Signal Processing: Processing and Perception Speech and Music, July 1999, John Wiley & Sons, ISBN: 0471351547
3. Lawrence Rabiner, Biing-Hwang Juang and B. Yegnanarayana, Fundamentals of Speech Recognition, Pearson, 1st Edition, 2009

4. Rabiner and Schafer, Digital Processing of Speech Signals, Pearson, 2003.
5. Thomas F. Quatieri, Discrete-Time Speech Signal Processing: Principles and Practice, Prentice Hall; ISBN: 013242942X; 1/e, 2001

IOE421 Large Language Models [3-0-0-3]

Course Prerequisite

- Probability and Statistics
- Machine Learning

Course Objectives

- Understand foundational concepts in natural language processing methods and strategies
- Provide a comprehensive understanding of transformer models, pre-training, and fine-tuning for modern Large Language Models (LLMs).
- Apply Transformers and Large Language models for building various NLP applications
- Develop awareness of the ethical implications, biases, and challenges in deploying responsible and fair LLM systems.

Course Objectives

- Describe the core NLP tasks, challenges, and the evolution from traditional methods to neural approaches in NLP.
- Analyze the architecture and operational principles of transformer-based models
- Apply pre-trained LLMs for practical tasks such as text classification, summarization, and text generation
- Evaluate the ethical and societal implications of deploying LLMs, focusing on bias, fairness, and responsible AI practices.

Syllabus

Introduction to Natural Language Processing: overview of NLP tasks, challenges, evolution. Review of Deep Neural networks-feedforward network, Backpropagation, RNN, Neural Language Models: Word Embedding. Seq2seq models: Encoder-decoder, Attention mechanism

Transformers and Large Language Models: Tokens and Embedding, Transformer Blocks, Self attention, Masked attention, Encoder only models (BERT), Decoder only models (GPT), Pre-training and Fine tuning , Applications of pre-trained models for Text classification, Summarization, Text generation

LLM Techniques and Applications: Decoding strategies, Prompt Engineering: In-Context Learning, and Instruct Tuning, Retrieval augmented generation, Alignment and PEFT, Responsible and Ethical LLMs: Bias and fairness, Guardrails and jailbreaking,

Lab Practice

- Basics of text processing: Processing Raw Text
- Text classification
- Sentiment Analysis
- Word embedding
- Transformers using huggingface/Langchain
- Fine Tuning a transformer model
- Prompt Engineering
- Retrieval Augmented Generation

Textbooks/References

1. Alammur, Jay, and Maarten Grootendorst. Hands-On Large Language Models: Language Understanding and Generation. O'Reilly Media, Inc., 2024.
2. Jurafsky, Daniel, and James H. Martin. "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition.", 3rd Edition, 2024.
3. Amaratunga, Thimira. "Understanding large language models: learning their underlying concepts and technologies." , Apress Media, 2023.
4. Vaswani, A. "Attention is all you need." Advances in Neural Information Processing Systems (2017).

HUMANITIES ELECTIVE COURSES

IHSXXX Introduction to Sociology

Course Objectives

- To understand the relationship between Science, Technology, Society and Culture.
- To familiarize with the social processes and dynamics of scientific knowledge and developments in Social Sciences
- To introduce Sociology as a science of society vis-à-vis Science and Technology.
- To know the fundamental social processes and institutions such as socialization, marriage, family, caste, gender etc.
- To discuss some of the important issues and problems in India today

Course Outcomes

- The relationship between Science, Technology, Society and Culture.
- Social processes and dynamics of scientific knowledge and developments in Social Sciences
- Sociology as a science of society vis-à-vis Science and Technology
- The fundamental social processes and institutions of socialization, marriage, family, caste and gender.

- Some of the important issues and problems in India today, such as Communalism, Diversity and Pluralism, Fundamentalism, Secularism, Globalization, Liberalization, Information Communication Revolution etc.

Syllabus

Sociology & Social Sciences, Science, Social Science and Society, Scientific Study of Society, Sociology - Science of Society; ,Approaches & Methods: Macro & Micro - Object & Subject, Objectivist & Interpretivist

Society, Social Structure & Culture ,Nature & Culture: Science and Society Interface ,Society and social structure, Culture,Socialization, Social Institutions - Marriage, Family, Caste, Class & Gender

India today - major issues and challenges, Diversity & Pluralism, Secularism & Communalism, Globalization & Consumer Culture, Information Communication Revolution - New media

Textbooks/References

1. Haralambos, Michael with Robin Heald. Sociology - Themes and Perspectives. NewDelhi : Oxford University Press, 1980.
2. Mohanty, Manoranjan, Class, Caste, Gender. New Delhi: Sage Publications, 2004.
3. Perry and Perry, Contemporary Society: An Introduction to Social Science. London: Allyn & Bacon, 2008.
4. Bryman Alan, Social Research Methods, Oxford University Press, Oxford, 2005.
5. Srinivas M.N, Caste in India and Other Essays. Delhi: Asia Publishing House, 1962.
6. Patel, Tulasi. The Family in India: Structure and Practice. New Delhi: Sage, 2005.
7. Knott-Cetina, Karin and Michael Mulkay, Science Observed: Perspectives on the
8. Social Study of Science. London: Sage Publications, 1983.Merton, Robert K, The Sociology of Science. Chicago: Univ. of Chicago Press, 1973.
9. Bijker, W. E. , Social construction of technology', International Encyclopedia of the Social and Behavioural Sciences. N. Smelser and P. Bailes, eds. Amsterdam, 2001.
10. Sarah Franklin, "Science as Culture, Cultures of Science," Annual Reviews of Anthropology 1995.
11. Sharma, K.L, Indian Social Structure and Change. Jaipur: Rawat Publications, 2007.
12. Bruno Latour and Steve Woolgar, Laboratory Life: The Construction of Scientific Facts Princeton NJ: Princeton University Press, 1986.

IHSXXX Taxation and Human Development /Economics

Course Objectives

The course seeks to sensitize the students to concepts of income,

human development, government's role in development, governance functions and taxation. It aims at raising the capacity of non- economics students to appreciate the need to understand some of these concepts for good citizenship as well for becoming able technologists.

Syllabus

Taxation and Central Transfers, Introduction to National Income, Gross Domestic Product (GDP), Gross Domestic Product and Gross National Income(GNI),Three Approaches to GDP, System of National Accounts and Production Boundary, Household's non-market Production,Women, Work, and the Economy:, Macroeconomic Gains from Gender Equity, Measuring Government Output, Satellite Account,Tourism Satellite Account for India, GDP to Welfare Measures, Human Development Index ,Purchasing Power Parities, National Income and Quarterly estimates of Gross Domestic Product (mospi.nic.in).

Textbooks/References

1. Ann Chadeau () What Is Households' NonMarket Production Worth? OECD Economic Studies No. 18, Spring 1992.
2. Katrin Elborgh-Woytek et al., (2013) Women, Work, and the Economy: Macroeconomic Gains from Gender Equity, IMF SDN/13/10.
3. Lequiller, F. and D. Blades (2014), Understanding National Accounts: Second Edition, OECD Publishing. (<http://dx.doi.org/10.1787/9789264214637-en>)
4. UNDP, Human Development Report 1990 and 2016.

IHSXXX Introduction to Digital Humanities [1-0-0-1]

Course Description

Digital Humanities (DH) looks at how the study of humanities has been immensely transformed by digital technology. It is an emerging discipline that studies the changing methodologies and pedagogies in humanities study and research in the light of the growing intersection of humanities with digital technology. This approach has a new relevance in the context of COVID 19 pandemic that has led to a knockout blow to the traditional approaches in education. Digital Humanities has a two-way approach; the systematic use of digital resources in humanities and the humanistic analysis of the application of digital resources. The discipline requires a certain amount of acquaintance with computer technology and the course can be handled in collaboration with faculties from Computer Science department.

Course Objectives

- To introduce the basic concepts and tools in Digital Humanities
- To familiarize students with new forms of textuality and digital archiving
- To develop competency in digital writing
- To incorporate digital tools and methods to support their own core studies and to develop practical skills in electronic archiving, processing, editing, etc.

Course Outcomes

By the end of the course students will be able to:

- Critically discuss digital humanities projects in light of current methods and theoretical approaches to the field
- Apply principles of textual, editorial and communication theory to technical situations, so that they can produce high quality work in a digital environment

Syllabus

A history of the Humanities in an age of digital computing- Transitions in humanities- Development of digital humanities- Beginnings of the intersection of humanities with computing- Ethics and the digital community-Digital Libraries and Archiving- Digital Humanities in India- Significant Digital Humanities Projects - Intersections of Digital Humanities with web designing, software programming, technical writing, Game Studies, etc. Job prospects -Data Manager, Data Designer- Digital Curator-Cultural Designer- Information Architect- Data ScientistMeta-data Analyst- Related disciplines (Medical Humanities, Astrobiology, etc.

Textbooks/References

1. Exploring Digital Humanities in India: Pedagogies, Practices, and Institutional Possibilities Edited By Maya Dodd and Nidhi Kalra, Routledge India, 2020.
2. A Companion to Digital Humanities, Eds. Schreibman, Siemens, and Unsworth, Blackwell, 2004
3. Debates in the Digital Humanities, Ed. Matthew Gold, University of Minnesota Press, 2012
4. Shillingsburg, Peter. (2006) From Gutenberg to Google. Electronic Representations of Literary Texts. Paperback. Cambridge University Press.

IHSXXX Visual Communication [1-0-0-1]

Course Description

Visual Communication examines the production, reception, and interpretation of the aspects of culture that communicate through visual imagery. These include not only art, but photography, and commercial imagery (including advertisements, popular film, and television). The emergence of visual culture is closely linked to the development of visual technologies.

Course Objectives

- To understand the historical, physiological, psychological, perceptual, and cognitive aspects of visual communication
- To critically analyse the ethical implications of electronic and print media images
- To examine how visual culture is inextricably tied to marketing and consumerism in the modern world
- To explore the possibilities of digital marketing

Course Outcomes

By the end of the course students will be able to:

- Build a critical understanding of the relationship between images and their representation

- To express the student's own ideas with creativity, clarity, and critical thinking

Syllabus

Introduction to Visual Communication- the politics of visibility- the virtual 'real' in contemporary culture- Consumption of mass culture- Photography, Cinema, Advertisement, Web content, etc.- The Digital World: The Virtual Reality in Contemporary Culture Theories of Visuality and impact on visual communication

New perspectives in visual culture- virtual usurps in the internet and cyberspaces- The misuse of visual technologies

1. Lester, Paul Martin. Visual Communication Images with Messages 4th Edition. Belmont, CA: Wadsworth Publishing Company, 2006.
2. Jones, Steve. Virtual Culture: Identity and Communication in Cybersociety. New Delhi: Sage, 2002.

IHSXXX Science Fiction and Graphic Narratives [1-0-0-1]

Course Description

The course looks at the origins of science fiction and graphic narratives to understand how science and technology intersect with the world of fiction. This course is designed for a reader community who will approach these works from a critical perspective.

Course Objectives

- To familiarize students with the history, culture and origins of science fiction and graphic narratives
- To appreciate the technological elements in science fiction and graphic narratives
- To promote the close reading of science fiction and visual storytelling materials

Course Outcomes

By the end of the course students will be able to:

- Identify the relationship between science and society
- Write analytically about fictional works

Syllabus

Origins of science fiction- sub-genres and themes Utopia- Dystopia- Time travel- Cybernetics- Issues of gender, ecology, etc. in science fiction

Origins of graphic narratives- Recent trends Emergence of web comics- Print versus web comics comics and technology

Textbooks/References

1. Paul Gravett. Graphic Novels. New York, Collins, 2005.
2. Bould, Mark et al (ed) - The Routledge Companion to Science Fiction, Routledge, London & New York, 2009.

IHSXXX Professional Communication [1-0-0-1]

Course Description

The course would cover the important aspects of communication in the professional scenario.

Emphasis would be given to key language skills inevitable for a successful professional life.

Course Objectives

- To understand the importance of communication in the professional/corporate world
- To attain fluency in oral communication
- To attain proficiency in technical writing
- To develop positive body language
- To learn the etiquettes to be practiced while participating in seminars and group discussions
- To acquire interview skills
- To practice interpersonal skills for better relations with seniors, juniors, peers and stakeholders

Course Outcomes

By the end of the course students will be able to:

- Draft effective proposals, circulars, minutes, etc.
- Prepare an attractive resume.
- To apply for a job and participate in selection process.
- Overcome the fear of interview process and be ready to perform at different levels of the selection process.
- To project a good personal image and social etiquette so as to have a positive impact on the building of one's chosen career.

Syllabus

Basics of professional/business communication developing social communication skills- vocabulary in the professional scenario- relevance of non-verbal communication- body language- haptics- chronemics Technical writing- writing proposals- email etiquettes- basics of research writing- abstract preparation- drafting a resume- cover letter- digital profile- video profile

Interview skills- Group discussion

Textbooks/References

1. Tebeaux, Elizabeth, and Sam Dragga. The Essentials of Technical Communication. OUP, 2018.
2. Raman, Meenakshi and Sangeeta Sharma. Technical Communication: Principles and Practice. OUP, 2015.

IHSXXX Advanced Communication Skills [1-0-0-1]

Non-Credit Course

Course objectives

This course is designed to build the vocabulary and language proficiency that are appropriate for IT professionals. This course will enable them to recognize problems with listening style and help them use listening skills to meet various professional needs. Effective communication abilities are helpful in comprehending, developing, and presenting technical presentations. Active listening is necessary to grasp the context in order to solve problems. To eliminate uncertainty during team activities, writing abilities must be standardized and sound.

Course Outcomes

After successful completion of this course students are able to:

- Understand and use various listening and speaking etiquette in everyday communication.
- Successfully read, comprehend, and critically evaluate both technical and nontechnical materials.
- Adapt writing techniques that are successful for both non-technical and technical communication.

Syllabus

Listening: active and passive listening, intensive listening, developing effective listening skills, barriers to effective listening, listening to longer technical talks, listening to classroom lectures, listening to documentaries and making notes, TED talks. Exercises based on audio materials like radio and podcasts, listening to songs, practice and exercises.

Spoken: phonetics, Pronunciation practice: Problem sounds, MMFS (Multimedia Feedback System), stress, intonation, pitch, uses of a dictionary for pronunciation practice. Conversation practices.

Reading, comprehension and summarizing Reading styles, speed, valuation, critical reading, SQ3R method, PQRS method - Speed reading Comprehension: techniques, understanding textbooks, marking and underlining - Note-taking: recognizing non-verbal cues.

Formal writing - Technical writing: differences between technical and literary style - Letter writing (formal, informal and semi-formal), job applications, minute preparation, CV preparation (differences between Bio-Data, CV and Resume), and reports - Elements of style - Common errors in writing, describing a process, Statement of Purpose, instructions, checklists.

Textbooks/References

1. John Seely, The Oxford Guide to Writing and Speaking, Oxford University Press, New Delhi, 2004.
2. Turton, N.D and Heaton, J.B, Dictionary of Common Errors, Addison Wesley Longman Ltd., Indian reprint 1998.
3. Anderson, P.V, Technical Communication, Thomson Wadsworth, Sixth Edition, New Delhi, 2007.

IHSXXX Soft Skills and Professional Ethics[1-0-0-1]

Course Objectives

This course is designed to improve students' soft skills and make them aware of work ethics. Students should be able to take part in debates, group discussions, interviews, and presentations, making use of effective communication and presentation techniques. Since good communication skills help further one's career, every effort will be made to increase the students' awareness of appropriate professional techniques. It is also designed to develop their writing and communication skills. This course will give more focus to the practical application of each module.

Course Outcomes

After successful completion of this course students are able to:

- Develop soft skills
- Groom corporate habits
- Face interviews with confidence.
- Aware about the work ethics
- Create the right impression in Interviews
- Guidance for paper presentation

Syllabus

Soft Skills - SMART Goals, Teamwork, Autodidacticism, Time Management, Stress Management, Positive Attitude, SWOT Analysis

Interview Skills - Self-Awareness, Confidence, Resume Writing, Body Language, Interview FAQs, Rejections.

Group Discussions & Mock Interviews, Practical Writing Skills - Article writing, Paper presentation for IT students, Analytical and issue based essays, Report writing: basics of report writing, structure of a report, types of reports, references, bibliography -Referencing style (IEEE Format) - Plagiarism

Professional Ethics - Responsibilities and Rights of Professionals, Managing Multiple Roles, Public Relations

Modern-day Research and Study Skills - Search engines and how to get optimal results using language, repositories, forums such as Git Hub, Stack Exchange, OSS communities (MOOC, SWAYAM, NPTEL), and Quora

Textbooks/References

1. Stephen E. Lucas, The Art of Public Speaking, 10th Edition; McGraw Hill Education, 2012.
2. David F. Beer and David McMurrey, Guide to Writing as an Engineer, John Wiley. New York, 2004.
3. Goodheart-Willcox, Professional Communication, First Edition, 2017.

4. Gopalaswamy Ramesh and Mahadevan Ramesh, The Ace of Soft Skills: Attitude, Communication and Etiquette for Success, Pearson Education; 1 edition, 2013.
5. Samson et.al. English for Life-4, New Delhi: Cambridge UP.